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455. ON PL"UCKER'S MODELS OF CERTAIN QUARTIC SURFACES.

[From the *Philosophical Transactions*, vol. CLX. (for the year 1870), pp. 87–103. Received and Read January 13, 1870.]

[The article begins on page 300 of this Volume.]

There are four real lines $(y = 0,\ x = \gamma)$, $(y = 0,\ x = \delta)$, $(z = 0,\ x = \gamma')$, $(z = 0,\ x = \delta')$.

The surface consists of a central pillow-like portion, joined on by two conical points to an upper portion, and by two conical points to an under portion, the whole being included between the planes $x = \gamma'$, $x = \delta'$.

Mod. 1 is 13: the form of the equation is

$$\frac{y^2}{l^2(x-\gamma)(x-\delta)} + \frac{z^2}{l'^2(x-\gamma')(x-\delta')} + 1 = 0;$$

the values γ', δ' lying between γ, δ; the principal conics are one of them a hyperbola, the other an ellipse, the vertices (on the axis or line $y = 0$, $z = 0$) of the hyperbola lying between those of the ellipse.

The variable conic, for values of x between γ', δ', has two real vertices, or it is a hyperbola; for the values, say, from γ' to γ, and δ' to δ, there are four real vertices, or the conic is an ellipse; for values beyond the limits γ, δ, there are two real vertices, and the conic is a hyperbola. There are the four real lines $(y = 0,\ x = \gamma)$, $(y = 0,\ x = \delta)$, and $(z = 0,\ x = \gamma')$, $(z = 0,\ x = \delta')$. The surface consists of 8 portions joined to each other by 8 conical points, but the form can scarcely be explained by a description.

Mod. 8 is 32: the form of the equation is

$$\frac{y^2}{l^2(x-\gamma)(x-\delta)} + \frac{z^2}{l'^2(x-\gamma)(x-\delta)} = 1,$$

viz., the principal conics are each of them a line-pair, the variable conic is always an ellipse.

There are the two real nodal lines $(y = 0,\ x = \gamma)$ and $(z = 0,\ x = \gamma')$, each of these being in the neighbourhood of the axis and beyond certain limits acnodal crunodal, the surface is a scroll, being, in fact, the well-known surface which is the boundary of a small circular pencil of rays obliquely reflected, and consequently passing through two focal lines.

Mod. 4 is 34: the equation is

$$\frac{y^2}{l^2(x-\gamma)(x-\delta)^2} + \frac{z^2}{l'^2(x-\gamma')(x-\delta)} + 1 = 0,$$

where $x = \delta$ not intermediate between the values $x = \gamma$ and $x = \gamma'$, say the order is δ, γ, γ'. The surface is thus a cubic surface; the principal conics are ellipses having on the axis a common vertex at the point $x = \delta$, and the remaining two vertices on the same side of the last-mentioned one. The variable conic for values between δ and γ has four real vertices, or it is an ellipse; for values between γ and γ' two real vertices, or it is a hyperbola; and for values beyond the limits δ, γ' it is imaginary.

There are on the surface the two real lines $(y = 0,\ x = \gamma)$ and $(z = 0,\ x = \gamma')$. The surface consists of a finite portion joined on by two conical points to the remaining portion.

Mod. 3 is 40: the form of equation is

$$\frac{y^2}{l^2(x-\gamma)(x-\delta)} + \frac{z^2}{l'^2(x-\delta)(x-\delta')} + 1 = 0.$$

The surface is thus a cubic surface; the principal conics are, one of them an ellipse, the other a pair of imaginary lines intersecting on the ellipse; for values of x between γ and δ, the variable conic has thus two real vertices, and it is a hyperbola; for values beyond these limits it is imaginary, and the whole surface is thus included between the planes $x=\gamma$ and $x=\delta$. There are the two real lines $(y=0,\ x=\gamma)$ and $(z=0,\ x=\delta)$.

Taking $l^2 = l'^2 = 1$, the surface is

$$\frac{y^2}{(x-\gamma)(x-\delta)(x-\delta')} + \frac{z^2}{(x-\delta)(x-\delta')} = 0,$$

which is a particular case of the parabolic cyclide.

As already mentioned, I have not completely identified the remaining models 9 to 14, but I will say a few words about them.

The equatorial surfaces, not included in the preceding 78 cases, Pl"ucker distinguishes (vol. II. p. 363) as "gedrehte" or "tordirte," say as twisted equatorial surfaces; the equation of such a surface is

$$\begin{aligned} &by^2 + 2hyz + az^2 + ab - h^2 = 0,\\ &\text{where}\qquad b = Fx^2 - 2Rx + B,\\ &\qquad\qquad\quad a = Ex^2 + 2Ux + G,\\ &\qquad\qquad\quad h = Kx^2 - Ox - G \quad (\text{or in particular } = -Ox - G). \end{aligned}$$

Mod. 13 is such a surface, being a twisted form of model 2.

Mod. 9 and Mod. 14 belong, I think, to the case $a=0$; viz., the form of the equation is $by^2 + 2hyz - h^2 = 0$. The variable conic is a hyperbola, the direction of one of the asymptotes being constant (vol. II. p. 368).

There are moreover (p. 372) equatorial surfaces in which the variable conic is always a parabola, and where there are on the surface four real or imaginary lines.

Mods. 10, 11, and 12 seem to represent such surfaces.

456. NOTE ON THE DISCRIMINANT OF A BINARY QUANTIC.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. x. (1870), p. 23.]

It is well known that the discriminant of a binary quantic $(a, b, c, d, \ldots \between x, y)^n$ is of the form

$$Ma + Nb^2,$$

but it is further to be remarked that if $b = 0$, then the form is

$$a(Ma + Nc^2),$$

if $b = 0$, $c = 0$, the form is

$$a^2(Ma + Nd^2),$$

and so on, until only the lowest two coefficients are not put $= 0$. Or, what is the same thing, if in the discriminant of the original function we put $a = 0$, then the discriminant divides by b^2; if $b = 0$, the discriminant divides by a, and, omitting this factor, if we then write $a = 0$, it divides by c^2; if $b = 0$, $c = 0$, the discriminant divides by a^2, and omitting this factor, if we then write $a = 0$, it divides by d^2 and so on, until as before.

Thus if $b = 0$, the discriminant of $(a, 0, c, d, e \between x, y)^4$, divides by a, and omitting this factor it is

$$\begin{aligned} & ac^3 \\ & - 18ac^2d^2 \\ & + 54acd^2e \\ & - 27ad^4 \\ & + 81c^4e \\ & - 54c^3d^2, \end{aligned}$$

which for $a = 0$ has the factor c^2; if $b = 0$, $c = 0$, the discriminant of $(a, 0, 0, d, e \between x, y)^4$ has the factor a^2, and omitting this factor it is

$$\begin{aligned} & ae^3 \\ & - 27d^4, \end{aligned}$$

which for $a = 0$ has the factor d^2; the series of theorems here terminates, since the lowest two coefficients d, e are not to be put $= 0$.

457. ON THE QUARTIC SURFACES $(\asymp\asymp\asymp\ U,\ V,\ W)^2 = 0$.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. x. (1870), pp. 24–34.]

I PROPOSE to myself for investigation the quartic surfaces represented by an equation

$$(\asymp\asymp\asymp\ U,\ V,\ W)^2 = 0,$$

where U, V, W are quadric functions of the coordinates.

Such a surface has 8 nodes (conical points), viz., these are the points of intersection of the quadric surfaces $U = 0$, $V = 0$, $W = 0$. It is to be observed, that not every quartic surface with 8 nodes is included under the above form; in fact the equation of a quartic surface contains (homogeneously) 35 coefficients, or say 34 arbitrary parameters; in order that a given point may be a node, 4 conditions must be satisfied, and it is consequently possible to find a quartic surface having 8 given points as nodes (and having in its equation $(34 - 8 \cdot 4 =)$ 2 arbitrary parameters): but 8 given points are not in general the intersections of three quadric surfaces, and such a quartic surface is therefore not in general included under the above form. I think, however, that it may be assumed that the above form includes all the quartic surfaces having 8 nodes, points of intersection of three quadric surfaces. It will presently appear that, included in the form, we have surfaces where (instead of the 8 nodes) there is a nodal or cuspidal conic; and that these are the most general forms of such quartic surfaces.

A quartic surface has at most 16 nodes, and the general form with 8 nodes must admit of being particularised so that the surface shall acquire any number not exceeding 8 of additional nodes. This does not show, but it is probable, that the above special form with 8 nodes can be particularised so that the surface shall in like manner acquire any number not exceeding 8 of additional nodes. Similarly, a quartic surface with a nodal conic may have besides 1, 2, 3, or 4 nodes and it will be shown in the sequel how the form, particularised so as to give a nodal conic, may be further particularised so as to give the 1, 2, 3, or 4 nodes. So a quartic surface with a cuspidal conic may besides have 1 node, and it will be shown how the form, particularised so as to give a cuspidal conic, may be further particularised so as to give 1 node.

Starting from the equation $(\asymp\asymp\asymp\ U,\ V,\ W)^2 = 0$, we may, by substituting for U, V, W lineal functions of these expressions, transform the equation precisely in the manner of a conic, and therefore into any of the forms under which the equation of a conic can be exhibited; for instance, in the forms $aU^2 + bV^2 + cW^2 = 0$, $fVW + gWU + hUV = 0$, $UW - V^2 = 0$, &c. I attend at present only to the last-mentioned form $UW - V^2 = 0$, which, it thus appears, is equally general with the original form $(\asymp\asymp\asymp\ U,\ V,\ W)^2 = 0$.

The quartic surface

$$UW - V^2 = 0,$$

where U, V, W are any quadric functions of the coordinates, may be considered as the envelope of the quadric surface

$$(\asymp\asymp\asymp\ U,\ V,\ W \pitchfork \theta,\ 1)^2 = 0,$$

where θ is an arbitrary parameter. And it thus appears that it is very easy to reciprocate (in regard to any given quadric surface) the quartic surface. For the reciprocal of the quartic surface is clearly the envelope of the reciprocal of the variable quadric surface; this reciprocal is itself a quadric surface, and the reciprocal of the quartic surface is thus given in the same form as the original surface, viz., as the envelope of a quadric surface the equation whereof contains rationally the variable parameter θ; the equation of the reciprocal surface is consequently obtained by equating to zero the discriminant in regard to θ, of the equation of the reciprocal quadric surface.

It is to be observed that, inasmuch as the equation of the reciprocal quadric surface is of the third degree in the coefficients of the original quadric, it is in general of the degree 6 in the parameter; we have thus a sextic function of θ, the coefficients whereof are quadric functions of the coordinates; and the discriminant is a function of the order 10 in these coefficients, that is, of the order 20 in the coordinates. The reciprocal of the quartic surface is thus a surface of the order 20; this is right, for in a general quartic surface the order of the reciprocal surface $= 36$, and the 8 nodes reduce the order by 16; $36 - 16 = 20$.

In the equation $UW - V^2 = 0$, or say $V^2 - UW = 0$; if U reduce itself to the square of a linear function, $U = P^2$, the equation becomes $V^2 - P^2W = 0$, which is the general form of the quartic surface having the nodal conic $V = 0$, $P = 0$. And moreover, if, W be the product of this same linear function P by another linear function Q, $W = PQ$, then the form is $V^2 - P^2Q = 0$, which is the general form of the quartic surface having the cuspidal conic $V = 0$, $P = 0$.

Writing for greater convenience x, y in the place of P, Q respectively, we have the quartic

$$(AB) \qquad V^2 - x^2y = 0,$$

having the cuspidal conic $V = 0$, $x = 0$; and which has besides the conic of plane contact $V = 0$, $y = 0$. In virtue of the cuspidal conic the reciprocal surface should be of the order 6; and by the foregoing method of obtaining the equation of the reciprocal surface, I will verify that this is so.

To effect this as simply as possible, I fix the remaining coordinates z, w as follows. The line $x = 0$, $y = 0$ is not in general a tangent to the surface $V = 0$; it therefore meets this surface in two points, and we may take $z = 0$, $w = 0$ to be the equations of the tangent planes at these two points respectively to $V = 0$; we have thus

$$V = ax^2 + 2hxy + by^2 + 2nzw.$$

Introducing for convenience the numerical factor 2, and taking the equation of the surface to be

$$(ax^2 + 2hxy + by^2 + 2nzw)^2 - x^2y = 0,$$

this is the envelope of the quadric surface

$$\theta^2ax^2 + 2\theta(ax^2 + 2hxy + by^2 + 2nzw) + 2xy = 0,$$

which is a surface $(a, b, c, d, f, g, h, l, m, n \between x, y, z, w)^2 = 0$, where $a = \theta^2 + 2a\theta$, $b = 2b\theta$, $h = 2h\theta + 1$, $f = 2\theta n$, and where all the other coefficients vanish. Assuming, as usual, that the reciprocation is effected in regard to the surface $x^2 + y^2 + z^2 + w^2 = 0$, the general equation is

$$\begin{aligned}
&x^2 \cdot \Delta(bc - f^2) - cm^2 - bn^2 + 2fmn \\
&+ y^2 \cdot \Delta(ca - g^2) - an^2 - cl^2 + 2gnl \\
&+ z^2 \cdot \Delta(ab - h^2) - bl^2 - am^2 + 2hlm \\
&+ w^2 \cdot \Delta abc - af^2 - bg^2 - ch^2 + 2fgh \\
&+ 2yz \cdot \Delta(gh - af) + l^2f + amn - hnl - glm \\
&+ 2zx \cdot \Delta(hf - bg) + m^2g + bnl - fhm - hmn \\
&+ 2xy \cdot \Delta(fg - ch) + n^2h + clm - gmn - fnl \\
&+ 2xw \cdot -\Delta\big[(bc - f^2) - n(hf - bg) - m(fg - ch)\big] \\
&+ 2yw \cdot -\Delta\big[(ca - g^2) - l(fg - ch) - n(gh - af)\big] \\
&+ 2zw \cdot -\Delta\big[(ab - h^2) - m(gh - af) - l(hf - bg)\big],
\end{aligned}$$

(I write down this general result as it will be useful for reference in other cases) in the present case this becomes simply

$$x^2 \cdot -bn^2 + y^2 \cdot -an^2 + 2z^2 \cdot nh^2 + 2zw \cdot (-nab + nh^2) = 0,$$

where a, b, n, h have the foregoing values; the equation is thus only of the order 4 in regard to θ; but it in fact divides by $n(=2\theta n)$ and thus reduces itself to the third order, viz. it becomes

$$n(bx^2+ay^2)-2h^2xy+2(ab-h^2)zw=0,$$

or, substituting for a, b, n, h their values, this is

$$x^2\cdot 4bn\theta^2+y^2\cdot 2n(2n\theta^2+4ab\theta)+2zw\cdot(2b\theta^2+4ab\theta)$$
$$-2(xy+zw)(2\theta h+1)^2=0,$$

say this is

$$(A,B,C,D \between \theta,1)^2=0,$$

where A, B, C, D are each of them a quadric function of the coordinates; it being observed that C and D are respectively numerical multiples of the same function $(xy+zw)$. Hence equating the discriminant to zero, we have

$$A^2D^2+4C^3+4B^2D-3B^2C^2-6ABCD=0,$$

which equation, inasmuch as every term contains either C or D as a factor, divides by $xy+zw$, and thus becomes an equation of the order 6 in the coordinates: that is the order of the reciprocal surface is $=6$. Multiplying by $\frac{4}{3}$ to avoid fractions, the actual values of A, B, C, D are

$$\begin{aligned}
A &= \tfrac{4}{3}(by^2+2hzw),\\
B &= 2\{h(ax^2+by^2)+2ahzw-2h^2(xy+zw)\},\\
C &= -4h(xy+zw),\\
D &= -\tfrac{4}{3}(xy+zw),
\end{aligned}$$

or say $A=\frac{4}{3}\alpha$, $B=2\beta$, $C=-4h\gamma$, $D=-\frac{4}{3}\gamma$; where $\gamma=xy+zw$; substituting these values and omitting the factor $\frac{4}{3}\gamma$, the equation is

$$27\alpha^2\gamma^2-256h^3\alpha\gamma^2-32\beta^3-64h\beta^2\gamma-144h\alpha\beta^2\gamma=0,$$

which is an equation of the form $(\mathrel{\approx\approx}\ \alpha,\beta,\gamma)^2=0$. The sextic surface has thus singular points $\alpha=0$, $\beta=0$, $\gamma=0$, viz. these are the two points $(x=0,\ y=0,\ z=0)$, $(x=0,\ y=0,\ w=0)$ each four times. The further discussion of the sextic surface is reserved for another occasion.

I do not at present attempt to enumerate the particular cases of the surface $V^2-x^2y=0$, but content myself with the discussion of a particular case in which the order of the reciprocal surface is $=3$. Suppose that $V=0$ is a cone, $y=0$ a tangent plane to the cone (so that the conic $y=0$, $V=0$ breaks up into a line twice repeated), $x=0$ an arbitrary plane (so that we have still the proper cuspidal conic $x=0$, $V=0$). Any other tangent plane of the cone may be taken for the plane $z=0$; the plane containing the lines of contact of the two tangent planes for the plane $w=0$; the equation of the conic then is $V=dw^2+2fyz=0$; and the equation of the surface is

$$(AB)\qquad (dw^2+2fyz)^2-x^2y=0.$$

For convenience of comparison, I change x, y, w, z into y, w, z, x, and assign numerical values to the coefficients, writing the equation under the form

$$(AB)\qquad 27(4zw+x^2)^2-64y^2w=0.$$

The quartic is here the envelope of the quadric surface

$$\theta^2\cdot 4yw+2\theta\cdot 9(4zw+x^2)+12y^2=0,$$

viz., comparing with the general form

$$(a, b, c, d, f, g, h, l, m, n \between x, y, z, w)^2 = 0,$$

we have $b = 12$, $c = 9\theta$, $l = 18\theta$, $m = 2\theta^2$, all the other coefficients vanishing. The reciprocal equation is

$$x^2(-cm^2) + y^2(-cl^2) + z^2(-bl^2) + 2xw \cdot dlm + 2yz \cdot (-lbc) = 0,$$

or substituting for b, c, l, m their values, this is found to be

$$\theta(\theta x - 9y)^2 + 108(z^2 + xw) = 0.$$

Representing this by $(A, B, C, D \between \theta, 1)^2 = 0$, the discriminant in regard to θ would, in virtue of the values of A, B, C, contain D as a factor; the reason of this appears from the original form; in fact, forming the derived equation in regard to θ, this is found to be $(\theta x - 9y)(\theta x - 3y) = 0$; the value $\theta x - 9y = 0$ gives as a factor of the discriminant $z^2 - 8xw$; the value $\theta x - 3y = 0$ gives $3y(-6y)^2 + 108x(z^2 + xw)$, that is the factor $y^2 + x(z^2 + xw)$; the complete value of the discriminant as obtained by substitution of the values of A, B, C, D being $x^2\{z^2 + xy\}\{y^2 + x(z^2 + xw)\}$; the equation of the reciprocal surface is

$$y^2 + x(z^2 + xw) = 0,$$

viz. this is a cubic surface. Schläfli's Prof. Case xx., having a uniplanar point $x = 0$, $y = 0$, $z = 0$ reducing the class by 8, and so giving a reciprocal surface of the order $(12 - 8 =)\ 4$, viz. the surface $27(4xw + z^2)^2 - 64y^2w = 0$. See the Memoir, Schläfli, "On the distribution of surfaces of the third order into species in reference to the absence or presence of singular points and the reality of their lines," *Phil. Trans.*, vol. CLIII, (1853), pp. 193–241.

I pass to the case of a surface

$$V^2 - P^2U = 0,$$

having a nodal conic $V = 0$, $P = 0$, but not having in general any nodes. And I propose to show how the constants may be determined so that the surface shall have 1, 2, 3, or 4 nodes. It is to be remarked that in the above equation the plane $P = 0$ is a determinate plane, but the quadric surface $V = 0$ is not a determinate quadric, we may in fact substitute for it the quadric $V + \lambda P^2 = 0$, writing the equation under the form

$$(V + \lambda P^2)^2 - P^2(U + 2\lambda V + \lambda^2 P^2),$$

so that we may without loss of generality, by means of the disposable constant λ, subject the surface $V = 0$ to any single condition; for instance, take it to be a cone, or to pass through a given point, &c.

Taking the planes $x = 0$, $y = 0$, $z = 0$, $w = 0$ to be arbitrary planes, the implicit constant factors in these equations may be determined in such wise that the equation of the given plane $P = 0$ shall be $x + y + z + w = 0$. The equation of the surface will then be

$$\{(a, b, c, d, f, g, h, l, m, n \between x, y, z, w)^2\}^2 = (x + y + z + w)^2 \cdot (a', b', c', d', f', g', h', l', m', n' \between x, y, z, w)^2,$$

and we may assume that the node or nodes (if any) lie at a vertex or vertices of the tetrahedron $x = 0$, $y = 0$, $z = 0$, $w = 0$, say at the points A, B, C, D. The conditions for a node at each of these points are at once found to be

Node at A,	Node at B,	Node at C,	Node at D,
$2a^2 = a' + a'$	$2bh = h' + b'$	$2cg = g' + c'$	$2dl = l' + d'$
$2ah = h' + a'$	$2bf = f' + b'$	$2cf = f' + c'$	$2dm = m' + d'$
$2ag = g' + a'$	$2bh = f' + h'$	$2c^2 = c' + c'$	$2dn = n' + d'$
$2al = l' + a'$	$2bm = m' + b'$	$2cn = n' + c'$	$2d^2 = d' + d'$

The first set of equations gives

$$a' = a^2, \quad h' = 2ah - a^2, \quad g' = 2ag - a^2, \quad l' = 2al - a^2.$$

If the first and second sets are satisfied simultaneously, we have $2(a-b)h = a^2 - b^2$, that is $a = b$, or else $h = \frac{1}{2}(a+b)$; that is, the two sets may be satisfied in two different ways according as a and b are equal or unequal. Similarly the first, second, and third sets may be satisfied in three different ways and the four sets in five different ways according as there are or are not any equalities between a, b, c, and between a, b, c and d respectively. The several solutions are shown in the annexed table, viz., in the line I no set is satisfied; in the line II only the first set; in the lines III and IV the first and second sets; in the lines V to VII the first, second, and third sets; and in the lines VIII to XII the four sets.

Table of Solutions.

	a	b	c	d	f	g	h	l	m	n	a'	b'	c'	d'	f'	g'
I	a	b	c	d	f	g	h	l	m	n	a^2	b^2	c^2	d^2	$2bf-b^2$	$2ag-a^2$
II	$\overline{a}$	b	c	d	f	g	h	l	m	n	0	b^2	c^2	d^2	$2bf-b^2$	$2ag-a^2$
III	$\overline{a}$	$\overline{b}$	c	d	f	g	$\frac{1}{2}(a+b)$	l	m	n	0	0	c^2	d^2	$2cf-c^2$	$2cg-c^2$
IV	$\overline{a}$	$\overline{b}$	c	d	f	g	h	l	m	n	0	0	c^2	d^2	$2cf-c^2$	$2ag-a^2$
V	$\overline{a}$	$\overline{b}$	$\overline{c}$	d	$\frac{1}{2}(b+c)$	$\frac{1}{2}(a+c)$	$\frac{1}{2}(a+b)$	l	m	n	0	0	0	d^2	0	0
VI	$\overline{a}$	$\overline{b}$	$\overline{c}$	d	$\frac{1}{2}(b+c)$	$\frac{1}{2}(a+c)$	h	l	m	n	0	0	0	d^2	bc	ac
VII	$\overline{a}$	$\overline{b}$	$\overline{c}$	d	f	g	h	l	m	n	0	0	0	d^2	$2bf-b^2$	$2ag-a^2$
VIII	$\overline{a}$	$\overline{b}$	$\overline{c}$	$\overline{d}$	$\frac{1}{2}(b+c)$	$\frac{1}{2}(a+c)$	$\frac{1}{2}(a+b)$	$\frac{1}{2}(a+d)$	$\frac{1}{2}(b+d)$	$\frac{1}{2}(c+d)$	0	0	0	0	0	0
IX	$\overline{a}$	$\overline{b}$	$\overline{c}$	$\overline{d}$	$\frac{1}{2}(b+c)$	$\frac{1}{2}(a+c)$	$\frac{1}{2}(a+b)$	l	m	n	0	0	0	0	0	0
X	$\overline{a}$	$\overline{b}$	$\overline{c}$	$\overline{d}$	$\frac{1}{2}(b+c)$	g	$\frac{1}{2}(a+b)$	$\frac{1}{2}(a+d)$	m	n	0	0	0	0	$2cg-c^2$	0
XI	$\overline{a}$	$\overline{b}$	$\overline{c}$	$\overline{d}$	f	$\frac{1}{2}(a+c)$	h	$\frac{1}{2}(a+d)$	$\frac{1}{2}(b+d)$	n	0	0	0	0	$2bf-b^2$	0
XII	$\overline{a}$	$\overline{b}$	$\overline{c}$	$\overline{d}$	f	g	h	l	m	n	0	0	0	0	$2bf-b^2$	$2ag-a^2$

I is the general case, $V^2 = P^2U$, of a quartic and a nodal conic but without nodes. (AC)

II is the case of a single node; writing, as without loss of generality we may do, $a = 0$, the equation is

$$[(0, b, c, d, f, g, h, l, m, n \pitchfork x, y, z, w)^2]^2 \\ = (x+y+z+w)^2 \cdot (b, c, d, m, l, f \pitchfork y, z, w)^2,$$

viz. the quadric $U = 0$ is here a cone having its vertex on the quadric $V = 0$. (AD)

III and IV are two cases each of them with two nodes, viz. III, the equation is

$$\{(ax+by)(x+y) + cz^2 + 2nzw + dw^2 + 2x(gz+lw) + 2y(fz+mw)\}^2 \\ = (x+y+z+w)^2 \{(ax+by)^2 + c'z^2 + 2n'zw + d'w^2 \\ + 2x[(2ag-a^2)z + (2al-a^2)w] \\ + 2y[(2bf-b^2)z + (2bm-b^2)w]\}, \tag{AE}$$

where it is to be observed that the line $z = 0$, $w = 0$ joining the two nodes $(y = 0, z = 0, w = 0)$ and $(x = 0, z = 0, w = 0)$ is a line on the surface. Writing, as we may do, $a = 0$, the equation assumes the more simple form

$$\{by(x+y) + cz^2 + 2nzw + d'w^2 + 2x(gz+lw) + 2y(fz+mw)\}^2 \\ = (x+y+z+w)^2 [b'y^2 + c'z^2 + 2n'zw + d'w^2 \\ + 2y\{(2bf-b^2)z + (2bm-b^2)w\}]. \tag{AE}$$

In IV the equation is

$$\begin{aligned}&\{a(x^2+y^2)+2hxy+cz^2+dw^2+2nzw+2x(gz+lw)+2y(fz+mw)\}^2\\ &\qquad=(x+y+z+w)^2\,[a^2(x^2+y^2)+2(2ah-a^2)xy+c'z^2+2n'zw+d'w^2\\ &\qquad\qquad+2x\{(2ag-a^2)z+(2al-a^2)w\}\\ &+2y\{(2af-a^2)z+(2am-a^2)w\}], \qquad (AF)\end{aligned}$$

where it will be observed that the line $z=0$, $w=0$ joining the two nodes is not a line on the surface.

Writing, as we may do, $a=0$, the equation becomes

$$\begin{aligned}&\{2hxy+cz^2+2nzw+dw^2+2x(gz+lw)+2y(fz+mw)\}^2\\ &=(x+y+z+w)^2\,(c'z^2+2h'zw+d'w^2), \qquad (AF)\end{aligned}$$

viz. the form is $F^2=P^2QR$, the quadric surface $U=0$ breaking up into the two planes $Q=0$, $R=0$; and the nodes being situate at the intersections of the line $Q=0$, $R=0$ with the surface $F=0$.

V, VI, VII are apparently cases with three nodes, but in fact VI is the only case of a proper quartic surface with three nodes. For in V the equation is

$$\begin{aligned}&\{(ax+by+cz)(x+y+z)+dw^2+2w(lx+my+nz)\}^2\\ &\qquad=(x+y+z+w)^2\,[(ax+by+cz)^2+d'w^2\\ &\qquad\qquad+2w\{(2al-a^2)x+(2bm-b^2)y+(2cn-c^2)z\}],\end{aligned}$$

which is satisfied by $w=0$, and the surface thus breaks up into the plane $w=0$ and a cubic surface.

And in VII the equation is

$$\begin{aligned}&[a(x^2+y^2+z^2)+dw^2+2fyz+2gzx+2hxy+2lxw+2myw+2nzw]^2\\ &\qquad=(x+y+z+w)^2\,[a^2(x^2+y^2+z^2)+d'w^2\\ &\qquad\quad+2\{(2af-a^2)yz+(2ag-a^2)zx+(2ah-a^2)xy\\ &\qquad\qquad+(2al-a^2)xw+(2am-a^2)yw+(2an-a^2)zw\}],\end{aligned}$$

which putting $a=0$ is

$$\begin{aligned}&\{dw^2+2fyz+2gzx+2hxy+2lxw+2myw+2nzw\}^2\\ &\qquad=(x+y+z+w)^2\,d'w^2,\end{aligned}$$

viz. this is a pair of quadric surfaces.

In the remaining case VI the equation is

$$\begin{aligned}&\{a(x^2+y^2)+2hxy+cz^2+dw^2+(a+c)(yz+zx)+2w(lx+my+nz)\}^2\\ &\qquad=(x+y+z+w)^2\,[a^2(x^2+y^2)+d'z^2+d'w^2+2cz(x+y)+2(2ah-a^2)xy\\ &+2w\{(2al-a^2)x+(2am-a^2)y+(2an-a^2)z\}], \qquad (AG)\end{aligned}$$

which putting therein $a=0$ is

$$\begin{aligned}&\{2hxy+dw^2+cz(x+y+z)+2w(lx+my+nz)\}^2\\ &=(x+y+z+w)^2\,\{c'z^2+d'w^2+2(2cn-c^2)zw\}, \qquad (AG)\end{aligned}$$

which is a surface having the nodes A, B, C; and it is to be observed that the lines CA, CB, but not the line AB, are lines on the surface.

IX, X, XI, XII are apparently cases with four nodes, but it is only XI which is a proper quartic with four nodes. In fact IX is

$$\begin{aligned}&\{(ax+ay+cz+dw)(x+y+z+w)+2(h-a)xy\}^2\\&\qquad=(x+y+z+w)^2\{(ax+ay+cz+dw)^2+4a(h-a)xy\},\end{aligned}$$

which is satisfied if $x=0$ or if $y=0$; that is, the surface breaks up into the two planes $x=0$, $y=0$, and a quadric surface.

X is

$$\begin{aligned}&\{a(x^2+y^2+z^2)+dw^2+2fyz+2gzx+2hxy\\&\qquad+(a+d)(xw+yw+zw)\}^2\\&\qquad=(x+y+z+w)^2\,[a^2(x^2+y^2+z^2)+d'w^2\\&\qquad+2(2af-a^2)yz+2(2ag-a^2)zx+2(2ah-a^2)xy\\&\qquad\qquad+2ad(xw+yw+zw)],\end{aligned}$$

which putting therein $a=0$ is

$$\begin{aligned}&\{dw(x+y+z+w)+2fyz+2gzx+2hxy\}^2\\&\qquad=(x+y+z+w)^2\,d'w^2,\end{aligned}$$

and thus breaks up into two quadrics.

And XII is

$$\begin{aligned}&[a^2(x^2+y^2+z^2+w^2)+2fyz+2gzx+2hxy+2lxw+2myw+2nzw]^2\\&\qquad=(x+y+z+w)^2\,[a^2(x^2+y^2+z^2+w^2)\\&\qquad+2(2af-a^2)yz+2(2ag-a^2)zx+2(2ah-a^2)xy\\&\qquad\qquad+2(2al-a^2)xw+2(2am-a^2)yw+2(2an-a^2)zw],\end{aligned}$$

which putting $a=0$ is

$$(2fyz+2gzx+2hxy+2lxw+2myw+2nzw)^2=0,$$

and is thus a quadric surface twice repeated.

There remains XI, and here the equation is

$$\begin{aligned}&\{(ax+ay+cz+cw)(x+y+z+w)+2(h-a)xy+2(n-c)zw\}^2\\&\qquad=(x+y+z+w)^2\{(ax+ay+cz+cw)^2+4a(h-a)xy+4c(n-c)zw\},\end{aligned}$$

or writing $h+a$, $n+c$ in place of h, n respectively, this is

$$\begin{aligned}&\{(ax+ay+cz+cw)(x+y+z+w)+2hxy+2nzw\}^2\\&=(x+y+z+w)^2\{(ax+ay+cz+cw)^2+4ahxy+4cnzw\},\qquad(AH)\end{aligned}$$

or putting herein $a=0$ it is

$$\{c(z+w)(x+y+z+w)+2hxy+2nzw\}^2-c(x+y+z+w)^2\{c(z+w)^2+4nzw\},\;(AH)$$

which may also be written

$$c(x+y+z+w)\{hxy(z+w)-nzw(x+y)\}+4hxy\cdot cnzw=0, (AH)$$

the equation of a quartic surface with the four nodes A, B, C, D; it is to be observed that the lines AC, AD, BC, BD are, the lines AB and CD are not, lines on the surface.

A more simple form may be given to the equation as follows; using the second of the above forms, multiplying the equation by 4, and writing therein

$$\begin{aligned} p &= \surd(c)\,(x+y+z+w), \\ q,\ r &= c(z+w)\pm 2\surd(cn)\,zw, \\ s,\ t &= c(x+y)\pm 2\surd(ah)\,xy, \end{aligned}$$

q, r, and s, t being the linear factors of the two quadric functions respectively, we have

$$qr-st=c(x+y+z+w)(-x-y+z+w)+4hxy+4nzw,$$

and thence

$$p^2+qr-st=2c(z+w)(x+y+z+w)+4hxy+4nzw,$$

wherefore the equation is

$$(p^2+qr-st)^2=4p^2qr, (AH)$$

or, what is the same thing,

$$p^2+\surd(qr)+\surd(st)=0, (AH)$$

where p, q, r, s, t are any linear functions of the coordinates; this is the equation of a quartic surface having the nodal conic $p=0$, $qr-st=0$; and the four nodes ($q=0,\ r=0,\ p^2-st=0$) and ($s=0,\ t=0,\ p^2-qr=0$). It includes the Cyclide, the equation of which may be written

$$b\xi^2=\surd\{(ax-eky+b'z)^2+b'^2z'^2\}+\surd\{(e'a-aky-b'z)^2+b'^2z'^2\}.$$

I remark that Prof. Kummer in his most valuable Memoir, "Ueber die Flächen vierten Grades auf welchen Schaaren von Kegelschnitten liegen," *Crelle*, t. LXVI. (1864), pp. 66–76, has considered several of the cases of a quartic surface with a nodal conic, viz. no node, (AC); a single node, (AD); two nodes (the case AF); and four nodes, (AH); but he has not considered two nodes, the case (AE); nor three nodes, (AG).

In reference to the general case of a quartic surface with a nodal conic, some most interesting properties have recently been obtained by Prof. Clebsch, see *Berl. Monatsh.*, April 30, 1868, where it is shown that there are on the surface 16 right lines forming 20 systems of double-fours, analogous in some respect to the 27 lines and 36 systems of double-sixes of a cubic surface.

458. ON THE ANHARMONIC-RATIO SEXTIC.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. x. (1870), pp. 56, 57.]

Mr Walker's equation is $A\Omega\Lambda^2 - 1 + iy + P(\lambda^2 - \lambda)^2 = 0$; changing the sign of λ, and also the numerical multipliers of I, A (so as to convert the discriminant equation into its standard form $\Delta = P^2 - 27J^2$), the equation is

$$4A(\lambda^3 + \lambda + 1)^2 - 27I^2(\lambda^2 + \lambda)^2 = 0.$$

I remark that this is most readily obtained as follows; writing

$$\begin{aligned} A &= (a-d)(b-c), \\ B &= (b-d)(c-a), \\ C &= (c-d)(a-b), \end{aligned}$$

then we have $A + B + C = 0$,

$$\begin{aligned} I &= -\tfrac{1}{4}(A^2 + B^2 + C^2) = -\tfrac{1}{2}(BC + CA + AB), \\ J &= \tfrac{1}{4\cdot 27}(B-C)(C-A)(A-B), \\ \surd(\Delta) &= \tfrac{1}{4\cdot 27}ABC, \end{aligned}$$

see my Fifth Memoir on Quantics, *Phil. Trans.*, vol. CXLVIII. (1858), pp. 429–460, [156]. And observe also, that in virtue of the relation $A + B + C = 0$, we have

$$-12I = A^2 + AB + B^2 = A^2 + AC + C^2 = B^2 + BC + C^2.$$

Hence writing

$$u = \frac{A\lambda + B}{\lambda + 1},$$

when λ has any one of the values

$$\frac{A}{B},\ \frac{B}{A},\ -\frac{A}{C},\ -\frac{C}{A},\ -\frac{B}{C},\ -\frac{C}{B},$$

we see that u assumes only the values A, B, C, and u is thus determined by the equation

$$u^3 - 12Iu - 16\surd(\Delta) = 0.$$

Eliminating u, we obtain

$$16\surd(\Delta)\left[\left(\frac{A\lambda + B}{\lambda + 1}\right)^3 - 12I\left(\frac{A\lambda + B}{\lambda + 1}\right) - 16\surd(\Delta)\right] = 0,$$

or, what is the same thing,

$$\left(\frac{A\lambda + B}{\lambda + 1}\right)^3 + \frac{1}{2}(A^2 + AB + B^2)\left(\frac{A\lambda + B}{\lambda + 1}\right) - \frac{1}{4\cdot 27}ABC = 0,$$

that is

$$4A(\lambda^2 + \lambda + 1)^2 - 27I^2\lambda^2(\lambda + 1)^2 = 0,$$

the required equation.

459. ON THE DOUBLE-SIXERS OF A CUBIC SURFACE.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. x. (1870), pp. 58–71.]

The 27 lines on a cubic surface include, and that in 36 different ways, a double-sixer; viz. a system of two sets of six lines

$$1,\ 2,\ 3,\ 4,\ 5,\ 6; \qquad 1',\ 2',\ 3',\ 4',\ 5',\ 6',$$

such that every line of the one set intersects all the non-corresponding lines of the other set, thus

	1’	2’	3’	4’	5’	6’
1	·	×	×	×	×	×
2	×	·	×	×	×	×
3	×	×	·	×	×	×
4	×	×	×	·	×	×
5	×	×	×	×	·	×
6	×	×	×	×	×	·

there being in all 30 intersections.

Any line, say 4, of the one set, intersects five lines $1'$, $2'$, $3'$, $5'$, $6'$ of the other set; and these six lines being given the double-sixer may be constructed; viz. (besides the line 4) we have a line 1 meeting the lines $2'$, $3'$, $5'$, $6'$; a line 2 meeting the lines $3'$, $5'$, $6'$, $1'$; a line 3 meeting the lines $5'$, $6'$, $1'$, $2'$; a line 5 meeting the lines $6'$, $1'$, $2'$, $3'$; and a line 6 meeting the lines $1'$, $2'$, $3'$, $5'$; and then the lines 1, 2, 3, 5, 6 are all of them met by a single line $4'$, which completes the system.

We may, if we please, consider the lines 4, 2 as given, and then $1'$, $3'$, $5'$, $6'$ will be any four lines each of them meeting the two given lines 4, 2; $2'$ will be any line meeting 4; and we have to determine a line $4'$ meeting 2, such that there may exist the lines 1, 3, 5, 6, completing the system as above. Or what is the same thing, we have a skew quadrilateral $1'$, 2, $3'$, 4; $5'$ and $6'$ meet 2 and 4; $2'$ meets 4, and $4'$ meets 2: 5 and 6 meet $1'$ and $3'$; 1 meets $3'$ and 3 meets $1'$; and the two sets $2'$, $4'$, $5'$, $6'$ and 1, 3, 5, 6 meet thus

	1	3	5	6
$2'$	·	×	×	×
$4'$	×	·	×	×
$5'$	×	×	·	×
$6'$	×	×	×	·

Hence, starting with the skew quadrilateral $1'2\,3'4$, and taking $x = 0$, $y = 0$, $z = 0$, $w = 0$ for the equations of the four planes $41'$, $1'2$, $23'$, $3'4$ respectively or what is the same thing $x = 0$, $y = 0$ for the equations of the line $1'$; $y = 0$, $z = 0$ for those of the line 2; $z = 0$, $w = 0$ for those of the line $3'$; and $w = 0$, $x = 0$ for those of the line 4; the several lines may be determined, each of them by means of its six coordinates, as follows

	a	b	c	f	g	h
$1'$						1
2				1		
$3'$			1			
4	1					
$2'$	A_2	B_2	C_2		G_2	H_2
$4'$		B_4	C_4	F_4	G_4	H_4
$5'$		A_5	C_5		G_5	H_5
$6'$		B_5	C_5		G_6	H_6
1	a_1	b_1	c_1	f_1	g_1	
3				f_3	g_3	
5				f_5	g_5	
6	a_6	b_6		f_6	g_6	

where

$$
\begin{aligned}
B_4A_2 + C_4H_2 &= 0,\\
B_4G_5 + C_5H_4 &= 0,\\
B_4G_6 + C_6H_4 &= 0,\\
B_5G_5 + C_5H_5 &= 0,\\
c_1f_3 + b_1g_1 &= 0,\\
f_5f_2 + b_5g_5 &= 0,\\
f_6f_2 + b_6g_6 &= 0,\\
a_1f_6 + b_1g_6 &= 0.
\end{aligned}
$$

The conditions in regard to the intersections of the lines $2'$, $4'$, $5'$, $6'$ and 1, 3, 5, 6, are formed by means of the diagram

	1	f_1	g_1	a_1	b_1	c_1	A_2	B_2	C_2	G_2	H_2	$2'$
3	f_3	g_3	A_3	C_3	f_3		B_4	C_4	F_4	G_4	H_4	$4'$
5	f_5	g_5		a_5	b_5		B_5	C_5		G_5	H_5	$5'$
6						B_6	C_6			G_6	H_6	$6'$

viz. we have the equations

first set,

$$\begin{aligned} f_1A_2 + g_1B_5 + b_1G_5 + c_1H_5 &= 0, \\ g_1B_5 + a_1f_5 + b_1G_5 + c_1H_5 &= 0, \\ g_3B_5 + b_3G_5 + c_3H_5 &= 0, \\ g_3B_6 + b_3G_6 + c_3H_6 &= 0; \end{aligned}$$

second set,

$$\begin{aligned} f_1A_2 + g_1B_4 + h_1C_4 + b_1G_4 &= 0, \\ g_1B_4 + h_1C_4 + a_1F_4 + b_3G_4 &= 0, \\ g_3B_4 + h_3C_4 + b_3G_4 &= 0, \\ g_5B_4 + h_5C_4 + b_5G_4 &= 0; \end{aligned}$$

third set,

$$\begin{aligned} f_3A_2 + g_3B_5 + b_3G_5 &= 0, \\ g_3B_5 + a_3F_5 + b_3G_5 &= 0, \\ g_5B_5 + b_5G_5 &= 0; \end{aligned}$$

fourth set,

$$\begin{aligned} f_3A_2 + g_3B_6 + b_3G_6 &= 0, \\ g_3B_6 + a_3F_6 + b_3G_6 &= 0, \\ g_5B_6 + b_5G_6 &= 0; \end{aligned}$$

and it is to be shown, that taking as given the coordinates of $2'$, $5'$, $6'$, that is $(A_2, B_2, C_2, G_2, H_2)$, (B_5, C_5, G_5, H_5) and (B_6, C_6, G_6, H_6), we can find the coordinates of the remaining lines $4'$, 1, 3, 5, 6.

The first set of equations gives

$$(g_1,\ b_1,\ c_1) = \begin{vmatrix} B_5, & G_5, & H_5 \\ B_6, & G_6, & H_6 \end{vmatrix},$$

viz. g_1, b_1, c_1 are proportional, but as only the ratios are material, they may be taken equal, to the determinants $G_5H_6 - G_6H_5$, $H_5B_6 - H_6B_5$, $B_5G_6 - B_6G_5$. And then retaining g_1, b_1, c_1 to signify these values respectively, the first equation gives f_1A_2, and the second equation gives a_1F_5; multiplying together these values, and writing $a_1f_1 = -b_1g_1$, we find

$$(B_5B_6,\ G_5G_6,\ H_5H_6,\ G_5H_6 + G_6H_5,\ H_5B_6 + H_6B_5,\ B_5G_6 + B_6G_5 + A_2F_5 \pitchfork g_1,\ b_1,\ c_1)^2 = 0.$$

Proceeding in a similar manner with the second set of equations, we have first

$$(h_3 = G_4B_5 - C_4B_5 = -C_4),$$

(observe $h_3 = G_4B_5 - C_4B_5 = -C_4$), and then

$$(B_5B_6,\ C_5G_6,\ G_5G_6,\ C_5G_6 + C_6G_5,\ G_5B_6 + G_6B_5 + A_4F_5,\ B_5C_6 + B_6C_5 \pitchfork g_3,\ h_3,\ b_3)^2 = 0.$$

The third set gives more simply

$$g_5^2B_5B_6 + g_5A_2(-B_5G_6 + B_6G_5 + A_2F_5) + b_5G_5G_6 = 0,$$

or since

$$g_5 : b_5 = G_6 : -B_6,$$

this is

$$G_6^2B_5B_6 - G_6B_5(B_5G_6 + B_6G_5 + A_2F_5) + B_5G_5G_6 = 0,$$

and similarly, the fourth set gives

$$g_5^2B_5B_6 + g_5A_2(B_5G_6 + B_6G_5 + A_2F_5) + h_5G_5G_6 = 0,$$

or since $g_5 : b_5 = G_6 : -B_6$, this is

$$G_6^2B_5B_6 - G_6B_5(B_5G_6 + B_6G_5 + A_2F_5) + B_5G_5G_6 = 0 :$$

and these last two results lead to the values of the ratios of B_4B_5, $B_4G_5 + B_5G_4 + A_4F_5$, G_4G_5; viz. these are proportional to expressions containing the common factor $B_5G_6 - B_6G_5$, and omitting this common factor, and taking them equal instead of merely proportional to the resulting expressions (which is allowable, since the absolute values are not material), we have

$$B_4B_5,\ B_4G_5 + B_5G_4 + A_4F_5,\ G_4G_5 = B_4B_5,\ B_4G_5 + B_5G_4,\ G_4G_5.$$

Returning to the result obtained from the first set of equations, this now becomes

$$(B_5B_6,\ G_5G_6,\ H_5H_6,\ G_5H_6 + G_6H_5,\ H_5B_6 + H_6B_5,\ B_5G_6 + B_6G_5 \pitchfork g_1,\ h_1,\ c_1)^2 = 0 :$$

but the terms containing g_1, b_1 are $(B_5g_1 + G_5b_1)(B_6g_1 + G_6b_1)$, viz. this is $= -H_5c_1 \cdot -H_6c_1$, that is $H_5H_6c_1^2$; the whole equation is thus divisible by c_1, and omitting this factor, it becomes

$$g_1(H_5B_6 + H_6B_5) + b_1(G_5H_6 + G_6H_5) + c_1(H_5H_6 + H_6H_5) = 0.$$

Proceeding in like manner with the result obtained from the second set of equations, this becomes

$$(B_5B_6,\ C_5G_6,\ G_5G_6,\ C_5G_6 + C_6G_5,\ B_5G_6 + B_6G_5,\ B_5C_6 + B_6C_5 \pitchfork g_3,\ h_3,\ b_3)^2 = 0,$$

where the terms containing g_3, b_3 are $(B_5g_3 + G_5b_3)(B_6g_3 + G_6b_3)$, viz. this is $-h_3G_6 \cdot -h_3G_5 = h_3^2G_5G_6$; the whole equation divides by h_3, and it then becomes

$$g_3(B_5G_6 + B_6G_5) + h_3(G_5G_6 + G_6G_5) + b_3(G_5G_6 + G_6G_5) = 0.$$

Considering B_4, G_4, F_4 as given by the equations

$$B_4B_5 = B_4B_5, \quad G_4G_5 = G_4G_5, \quad B_4G_5 + G_4B_5 + A_4F_4 = B_4G_5 + B_5G_4,$$

the equations last obtained determine the values of H_4 and C_4, viz. these equations may be written

$$(g_1B_5 + h_1C_5 + b_1G_5)C_4 + C_5(g_1B_4 + b_1G_4) + h_1G_5C_4 = 0;$$

but in order that the values $(B_4, C_4, F_4, b_4, H_4)$ given by these five equations may belong to a line $(0, B_4, C_4, F_4, G_4, H_4)$, they must satisfy the equation

$$B_4G_4 + C_4H_4 = 0,$$

viz. in order to the existence of the line $4'$, this equation must be satisfied identically by the foregoing values; and I proceed to show that it is in fact thus satisfied.

Multiplying the values of C_4, H_4, and writing $C_1H_2 = -B_1G_2$, the identity to be verified is

$$\begin{aligned}(g_1 \cdot -B_5 + b_1G_5 + c_1H_5)(g_1 \cdot B_6 + h_1C_6 + b_5G_6 \cdot -B_4 \cdot G_4)&\\ + \{H_5(g_1B_4 + b_1G_4) + (h_1H_5H_4\}\{C_5(g_1B_4 + b_1G_4) + h_1C_5C_4\} &= 0.\end{aligned}$$

The first line includes the terms

$$\{g_1g_3B_5^2 + b_1b_3G_5^2 + (b_1g_3 + b_3g_1)B_5G_5 + c_1C_5H_5\}B_4G_4,$$

which, writing $C_1H_2 = -B_1G_2$ and $B_2B_3 = B_4B_5$, $G_2G_3 = G_4G_5$, are

$$= g_1g_3B_4G_4B_5B_6 + b_1b_3G_4B_4G_5G_6 + (b_1g_3 + g_1b_3 - c_1h_3)B_4B_5G_4G_6.$$

The second line includes the terms

$$C_4H_4(g_1B_5 + b_1G_5)(g_3B_6 + h_3C_6) + c_1h_3C_4H_4C_5H_5,$$

which, reducing in like manner, are

$$= -g_1g_3G_4B_4B_5B_6 - b_1b_3B_4G_4G_5G_6 - (b_1g_3 + g_1b_3 - c_1h_3)B_4B_5G_4G_6,$$

and these are together

$$= (g_1g_3B_4B_5 - b_1b_3G_4G_6)(B_4G_5 - B_5G_4).$$

The remaining terms from the first line are at once reduced to

$$(g_1h_3C_5C_6 + c_1h_3H_5G_6)B_4B_5 + (b_1h_3G_5B_6 + c_1h_3H_5B_4)G_4G_6,$$

and those from the second line are

$$G_4C_4\,H_5(g_1B_5 + b_1G_5) + H_5H_4\,C_5(c_1g_1B_5 + c_1b_1G_5).$$

Hence, attending to the relation $C_3 = -h_3$, and collecting and arranging, the equation to be verified is

$$\begin{aligned}(g_1g_3B_4B_5 - b_1b_3G_4G_6)(B_4G_5 - B_5G_4)&\\ + h_3G_4G_6(g_3B_5B_6 - b_3H_5H_6)&\\ + h_3H_5H_6(g_1C_5C_6 - b_1G_5G_6)&\\ + h_3C_5B_5(b_3G_5G_6 - g_3H_5H_6)&\\ + h_3H_5G_5(b_1 \cdot G_4G_5 - g_1 \cdot B_4B_5) &= 0.\end{aligned}$$

But we have

$$\begin{aligned} g_1B_5B_6 - h_3H_5H_6 &= B_5B_6(G_5H_6 - G_6H_5) - H_5H_6(B_5G_6 - B_6G_5)\\ &= B_5H_5(B_6G_6 + C_6H_6) - B_6H_6(B_5G_5 + C_5H_5) = 0,\end{aligned}$$

and similarly

$$\begin{aligned} g_3G_5G_6 - h_3G_5G_6 = 0,\\ h_3G_5G_6 - g_3H_5H_6 = 0,\\ b_3C_5C_6 - g_3B_4B_5 = 0.\end{aligned}$$

Moreover, writing $g_1B_4B_5 = h_3H_5H_6$, we have

$$\begin{aligned} &g_1g_3B_4B_5 - b_1b_3G_4G_6\\ &= b_1(g_3H_5H_6 - b_3G_5G_6) = 0.\end{aligned}$$

and the five terms of the equation in question thus separately vanish; and the equation is consequently verified.

We may collect the results as follows:

Data are lines $1'$, 2, $3'$, 4, $2'$, $5'$, $6'$;

and then, for the remaining lines, 1, 3, 5, 6, $4'$ the coordinates are as follows:

For $4'$,

$$B_4B_5 = B_4B_5, \quad G_4G_5 = G_4G_5, \quad A_4F_4 = B_4G_5 + B_5G_4 - B_4G_5 - B_5G_4, \quad A_4 = 0,$$

$$\begin{vmatrix} B_5, & G_5, & H_5\\ B_6, & G_6, & H_6\end{vmatrix} H_4B_5, \quad H_4G_5, \quad H_4H_5 = 0,$$

$$\begin{vmatrix} B_5, & G_5, & H_5\\ B_6, & G_6, & H_6\end{vmatrix} B_4, \quad G_4, \quad H_4$$

$$\begin{vmatrix} B_4, & C_4, & G_4\\ c_4 + C_5B_4, & G_4c_4, & C_5G_4\end{vmatrix} = 0,$$

$$\begin{vmatrix} B_4, & G_4, & G_4\\ B_5, & G_5, & G_5\end{vmatrix}$$

$$\begin{vmatrix} B_5, & C_5, & G_5\\ B_6, & C_6, & G_6\end{vmatrix} \quad (B_4G_4 + C_4H_4 = 0,\ \text{identity}).$$

For 1,

$$(g_1, b_1, c_1) = \begin{vmatrix} B_5, & G_5, & H_5\\ B_6, & G_6, & H_6\end{vmatrix}, \quad f_1 = \frac{1}{A_2}\cdot\begin{vmatrix} B_5, & G_5, & H_5\\ B_6, & G_6, & H_6\end{vmatrix}, \quad F_1a_1 = \begin{vmatrix} B_4, & G_4, & H_4\\ B_5, & G_5, & H_5\end{vmatrix}, \quad g_1 = 0,$$

$$\begin{vmatrix} B_5, & G_5, & H_5\\ B_6, & G_6, & H_6\end{vmatrix}\begin{vmatrix} B_4, & G_4, & H_4\\ B_5, & G_5, & H_5\end{vmatrix}$$

$(a_1f_1 + b_1g_1 = 0,\ \text{identity})$.

For 3,

$$(g_3,\ h_3,\ b_3) = \begin{vmatrix} A_3, & G_5, & G_4\\ B_5, & C_5, & G_5\end{vmatrix}, \quad A_3f_3 = -\begin{vmatrix} A_3, & C_5, & G_4\\ B_5, & C_5, & G_5\end{vmatrix}, \quad F_3a_3 = \begin{vmatrix} B_4, & G_4, & G_4\\ B_5, & G_5, & G_5\end{vmatrix}, \quad C_3 = 0,$$

$$\begin{vmatrix} B_4, & G_4, & G_4\\ B_5, & G_5, & G_5\end{vmatrix}$$

$(a_3f_3 + b_3g_3 = 0,\ \text{identity})$.

For 5,

$$g_5 = G_6, \quad b_5 = -B_6, \quad A_2 f_5 = B_5 G_6 - B_6 G_5, \quad F_5 a_5 = B_5 G_6 - B_6 G_5, \quad c_5 = 0, \quad h_5 = 0,$$

$(a_5 f_5 + b_5 g_5 = 0,\ \text{identity})$.

For 6,

$$g_6 = G_5, \quad b_6 = -B_5, \quad A_2 f_6 = B_5 G_6 - B_6 G_5, \quad F_6 a_6 = B_5 G_6 - B_6 G_5, \quad c_6 = 0, \quad h_6 = 0;$$

and, for actual calculation, it is convenient to remark that as only the ratios are material, a set of six coordinates may be multiplied or divided by any common number at pleasure.

But these results may be further reduced. Writing

$$(g_1,\ h_1,\ c_1) = \begin{vmatrix} -a_3, & G_4, & G_5 \\ B_5, & C_5, & G_5 \end{vmatrix},$$

we have

$$-(g_1 B_5 + h_1 C_5 + b_1 G_5) = \begin{vmatrix} C_5, & G_5 \\ c_4, & G_4 \end{vmatrix},$$

$$(g_1 B_5 + h_1 C_5 + b_1 G_5) + h_1 C_5 C_4 = \frac{1}{G_5 C_4}\{g_1 B_5 B_4 G_4 + h_1 G_5 G_4 B_4\} + h_1 C_5 C_4.$$

But

$$\begin{aligned} &g_1 B_5 B_4 G_4 + h_1 G_5 G_4 B_4 \\ &= \frac{B_4 G_4}{G_5 C_4}\{g_1 B_5 G_5 C_4 (G_4 H_5 - G_5 H_4) + B_5 G_5 H_4 (G_5 B_4 - G_4 B_5)\} \\ &= C_5 G_4 \{B_4 (G_5 H_6 - G_6 H_5) + G_4 (B_5 H_6 - B_6 H_5)\} \\ &= C_5 G_4 (B_4 g_1 + G_4 h_1), \end{aligned}$$

since

$$(g_1,\ h_1,\ c_1) = \begin{vmatrix} B_5, & G_5, & H_5 \\ B_6, & G_6, & H_6 \end{vmatrix},$$

the equation obtained is thus

$$g_1 B_4 + h_1 G_4 = \frac{C_5 G_4}{G_5 C_4}(g_1 B_5 + b_1 G_5).$$

We then have

$$\begin{aligned} -(g_1 B_5 + h_1 C_5 + b_1 G_5)\, G_4 &= \frac{C_5 G_4}{G_5 C_4}[C_5 (g_1 B_5 + b_1 G_5) + h_1 B_5 G_5] \\ &= \frac{C_5 G_4}{G_5 C_4}[C_5 (g_1 B_5 + b_1 G_5) + c_1 C_5 H_5] \\ &= \frac{C_5 G_4}{G_5 C_4}(g_1 B_5 + h_1 C_5 + c_1 H_5), \end{aligned}$$

that is

$$\frac{C_4}{H_4} = \frac{C_5 G_4}{G_5 C_4}\,\frac{g_1 B_5 + h_1 C_5 + c_1 H_5}{g_1 B_5 + b_1 G_5 + c_1 H_5}.$$

and in like manner

$$\begin{aligned}
-(g_1B_5+b_1G_5+c_1H_5)\,H_4 &= \frac{C_5G_4}{G_5C_4}\{g_1G_5B_5B_6+h_1B_5G_5G_6\}+c_1H_5H_6\\
&= \frac{G_5B_5B_6(G_5H_6-G_6H_5)}{G_5C_4}+\frac{B_5G_5G_6(B_5H_6-B_6H_5)}{G_5C_4}\\
&\quad + B_5G_5G_6\{H_5B_5-H_5G_5\}+B_5G_5H_5(B_5G_6-B_6G_5)\\
&= H_5H_6[B_5(C_5C_6-C_6G_5)+G_5(B_5C_6-B_6G_5)]\\
&= H_5H_6(B_5g_1+G_5h_1);
\end{aligned}$$

the equation obtained is thus

$$g_1B_5+h_1G_5=\frac{C_4H_4}{C_5G_4}(B_5g_1+G_5h_1)$$

and then

$$\begin{aligned}
-(g_1B_5+b_1G_5+c_1H_5)\,H_4 &= \frac{C_4H_4}{C_5G_4}\{H_5(B_5g_1+G_5h_1)+c_1B_5G_5\}\\
&= \frac{C_4H_4}{C_5G_4}[H_5(B_5g_1+G_5h_1)+h_1C_5H_5]\\
&= -\frac{C_4H_4}{C_5G_4}(B_5g_1+C_5h_1+G_5h_1),
\end{aligned}$$

that is

$$\frac{H_4}{F_4}=\frac{H_5H_6}{C_5}\,\frac{g_3B_5+h_3C_5+b_3G_5}{g_1B_5+b_1C_5+c_1H_5},$$

which values of C_4, H_4 satisfy, as they should do, the relation

$$B_4G_4+C_4H_4=0.$$

We have also

$$\begin{aligned}
A_4F_4 &= B_4G_5+B_5G_4-\frac{G_5B_5B_6}{G_4}-\frac{G_4B_4C_5}{B_5}\\
&= \frac{1}{B_5G_4}(G_4B_5-B_4G_5)(G_4B_5-B_5G_4),
\end{aligned}$$

which gives F_4.

Moreover

$$\begin{aligned}
-F_4a_4 = g_1B_5+h_1G_5+c_1H_5 &= \frac{C_4H_4}{C_5G_4}(g_1B_5+h_1G_5)\\
&= \frac{C_4H_4}{C_5G_4}(g_1B_5+h_1G_5)+c_1H_5\cdot\frac{C_4H_4}{C_5G_4}\\
&= \frac{H_5H_6}{C_5}\,\frac{(g_1B_5+b_1G_5)(g_1B_5+h_1G_5+c_1H_5)}{(g_1B_5+h_1G_5+b_1G_5)h_1H_5}\\
&= \frac{H_5H_6}{C_5H_4}\,\frac{(g_1B_5+b_1G_5+c_1H_5)\{(g_1B_5+h_1G_5)(g_1B_5+b_1G_5)+c_1h_1B_5G_5\}}{(g_1B_5+h_1G_5+b_1G_5)}
\end{aligned}$$

which combined with the foregoing value of F_4, gives a_4.

Again,

$$\begin{aligned}-F_3a_3 = g_3B_5 + h_3C_5 + b_3G_5 &= \frac{C_4H_4}{C_5G_4}(g_3B_5 + b_3G_5) = \frac{C_4H_4}{C_5G_4}\frac{g_3B_5 + h_3C_5 + b_3G_5}{c_1H_4}\\ &= \frac{C_4H_4}{C_5G_4}\frac{(g_3B_5 + b_3G_5)(g_3B_5 + h_3G_5) + c_1h_3B_5C_5}{C_5H_4(g_3B_5 + h_3C_5 + b_3G_5)},\end{aligned}$$

which combined with the foregoing value of F_3, gives a_3.

Write

$$\omega = B_5G_5 + B_6G_6 + C_5H_5 + G_5H_6,$$

we have

$$\begin{aligned}b_1g_1 &= (H_5B_6 - H_6B_5)(G_5H_6 - G_6H_5)\\ &= -H_5B_5G_6 - H_6B_6G_5 + H_5H_6(B_5G_6 + B_6G_5)\\ &= H_5H_6(C_5H_5 + C_6H_6 + B_5G_6 + B_6G_5),\end{aligned}$$

that is

$$b_1g_1 = H_5H_6\,\omega,$$

and similarly

$$\begin{aligned}b_3g_3 &= 0,\ a_3\,\omega,\\ b_1b_3 &= B_4B_5\,\omega,\\ g_1g_3 &= G_5G_6\,\omega,\\ h_1g_3 + b_3g_1 + C_1h_3 &= -(B_5G_6 + B_6G_5)\,\omega,\\ (b_1G_5 + g_1B_5)(h_3G_5 + g_3B_5) + c_1h_3B_5G_5 &= \{G_5^2B_4B_5 + B_5^2G_5G_6 - B_4B_5(B_5G_6 + B_6G_5)\}\,\omega\\ &= (G_5B_5 - B_5G_6)(G_5B_4 - B_5G_5)\,\omega,\end{aligned}$$

which last value is to be substituted for the left-hand function in the formulae for a_1 and a_3 respectively.

Whence, finally recollecting that

$$B_4G_4 + G_5H_5 = 0,\quad B_4G_4 + G_6H_6 = 0,\quad B_5G_5 + C_5H_5 = 0,$$

and

$$\omega = C_5H_5 + G_5H_6 + B_5G_5 + B_6G_6,$$

we have

$$\begin{vmatrix} B_5, & G_5, & H_5\\ B_6, & G_6, & H_6\end{vmatrix} = -\,(g_1B_5 + b_1G_5 + c_1H_5).$$

For 1

$$b_1g_1 = H_5H_6\,\omega,\qquad a_1 = \frac{A_2H_5H_6\,C_5}{g_1B_5 + h_1G_5 + c_1H_5},\qquad h_1 = 0.$$

For 3

$$(g_3,\ h_3,\ b_3) = \begin{vmatrix} B_4, & C_4, & G_4 \\ B_5, & C_5, & G_5 \end{vmatrix}, \qquad a_3 = \frac{A_2 B_5 G_5\, C_5}{g_5 B_5 + h_5 G_5 + b_3 G_5}, \qquad h_3 = 0,$$

where observe that $C_3 + h_3 = 0$.

For 5

$$h_5 = G_6, \quad b_5 = -B_6, \quad A_2 f_5 = \tfrac{1}{2}(G_5 B_6 - B_5 G_6), \qquad a_5 = \frac{A_2 B_5 G_6}{G_5 B_6 - B_5 G_5},$$

$$c_5 = 0, \quad h_5 = 0.$$

For 6

$$g_6 = G_5, \quad b_6 = -B_5, \quad f_6 = \tfrac{1}{2}(G_5 B_6 - B_5 G_6), \qquad a_6 = \frac{A_2 B_5 G_6}{G_5 B_6 - B_5 G_5},$$

$$c_6 = 0, \quad h_6 = 0.$$

For 4′

$$B_4 = C_5 G_4, \quad g_4 = \frac{g_1 B_5 + b_1 G_5 + c_1 H_5}{H_4\, g_1 B_5 + h_1 G_5 + b_1 G_5},$$

$$H_4 = \frac{H_5 H_6}{C_5}\,\frac{g_3 B_5 + h_3 G_5 + b_3 G_5}{g_1 B_5 + b_1 G_5 + c_1 H_5}.$$

I have thought it worth while to effect the numerical calculations for enabling the construction of a drawing or model. For this purpose taking X, Y, Z as ordinary rectangular coordinates, I write

$$\begin{aligned} x &= X + Y + Z - 10, \\ y &= X - Y + Z - 10, \\ z &= -X + Y + Z - 10, \\ w &= 7, \end{aligned}$$

that is, I take $1'$ and 2 to be lines in the plane of XY, defined by the equations $X+Y=10$ and $X-Y=-10$ respectively, and $3'$ and 4 to be lines in the plane of XZ defined by the equations $X-Z=-10$, $X+Z=10$ respectively. And I take $5'$ to be the line joining the points $(2,0,8)$ and $(-9,1,0)$; $6'$ the line joining the points $(3,0,7)$ and $(-8,2,0)$; $2'$ the line joining the points $(9,0,1)$ and $(-3,7,4)$. We calculate then for each of the lines the $xyzw$ coordinates of two points thereof; and thence the six coordinates of the line, viz.

	X	Y	Z	w	X	Y	Z	w	A	B	C	F	G	H
$5'$	0,	8,	-4,		-18,	0,	0,		0,	72,	144,	0,	8,	-4
									0,	18,	36,	0,	2,	-1
$6'$	0,	7,	-6,		-16,	0,	0,	2	0,	96,	112,	0,	14,	-12
									0,	48,	56,	0,	7,	-6
$2'$	0,	1,	-18,		-2,	4,	4,	7		76,	36,	2,	7,	-126
									76,	36,	2,	0,	7,	-126

and effecting the calculations for the remaining lines, we have

	A	B	C	F	G	H
$4'$				$\frac{9}{2}$		$\frac{3}{38}$
					$\frac{24}{-944}$	$\frac{59}{385}$
1			$\frac{380}{59}$		$\frac{30}{-5}$	$\frac{19}{15}$
3	127680	-720			$\frac{140}{19}$	-30
5		$\frac{304}{3}$	-48		$\frac{7}{19}$	
6	152	-18			$\frac{27}{38}$	

or reducing to integers, the values are

	B	C	F	G	H
$5'$	18	36		2	-1
$6'$	48	56		7	-6
$2'$	76	36	2	7	-126
$4'$		53808	-2116448	-531	4484
1	1444	13452	6726	10443	-1121
3	485184	-2736		3	532
5	5776	-912		21	133
6	5776	-2052		81	228

The line (a, b, c, f, g, h) is given as the intersection of any two of the four planes

$$\begin{aligned}(0,\ h,\ -g,\ a \pitchfork x, y, z, w) &= 0,\\ (-h,\ 0,\ f,\ b \pitchfork x, y, z, w) &= 0,\\ (g,\ -f,\ 0,\ c \pitchfork x, y, z, w) &= 0,\\ (-a,\ -b,\ -c,\ 0 \pitchfork x, y, z, w) &= 0.\end{aligned}$$

Or substituting for x, y, z, w the values $X+Y+Z-10$, $X-Y+Z-10$, $-X+Y+Z-10$, 7, these become

$$\begin{aligned}(-f-h,\ b+f-h,\ f-h,\ h-g \pitchfork X, Y, Z, 10) &= 0,\\ (g,\ c+g,\ g-f,\ -g \pitchfork X, Y, Z, 10) &= 0,\\ (c-a,\ -c-a,\ -c-a-b,\ c+a \pitchfork X, Y, Z, 10) &= 0,\end{aligned}$$

or, what is the same thing,

$$\begin{aligned}(-2g+a-c,\ 2g-f-h,\ a+b+c+f-h,\ -2g,\ -a-c \pitchfork X, Y, Z, 10) &= 0,\\ (-2g+f+h,\ -a-b-c-f+h,\ 2g-f-h,\ -f+h \pitchfork X, Y, Z, 10) &= 0.\end{aligned}$$

And substituting, we have the equations of the several lines, viz.:

$$\begin{array}{ll}
(1') & Z+Y=10, \quad Z=0;\\
(2) & -Z+Y=10, \quad Z=0;\\
(3') & -Z+X=10, \quad Y=0;\\
(4) & X+Z=10, \quad Y=0;\\
(5') & (\cdot \quad 40, \quad -4 \pitchfork X, Y, Z, 10) = 0,\\
 & \quad (-40, \quad \cdot \quad 55, \quad -36 \pitchfork X, Y, Z, 10) = 0,\\
 & \quad (-5, \quad -55, \quad \cdot \quad 4, \quad 36, \quad -1 \pitchfork X, Y, Z, 10) = 0;\\
(6') & (\cdot \quad -35, \quad 10, \quad -7 \pitchfork X, Y, Z, 10) = 0,\\
 & \quad (-35, \quad \cdot \quad 1, \quad 55, \quad -28 \pitchfork X, Y, Z, 10) = 0,\\
 & \quad (-10, \quad -55, \quad 7, \quad 28, \quad -3 \pitchfork X, Y, Z, 10) = 0.
\end{array}$$

$$
\begin{array}{ll}
(2') & (\cdot \quad -30, \quad 70, \quad -7 \pitchfork X, Y, Z, 10) = 0, \\
 & (30, \quad \cdot \quad 120, \quad -39 \pitchfork X, Y, Z, 10) = 0, \\
 & (-70, \quad -120, \quad \cdot \quad 63, \quad 7, \quad 39, \quad -63 \pitchfork X, Y, Z, 10) = 0; \\
(4') & (\cdot \quad -2107480, \quad 9385, \quad -8968 \pitchfork X, Y, Z, 10) = 0, \\
 & (2107480, \quad \cdot \quad -2063, \quad 2116448 \pitchfork X, Y, Z, 10) = 0, \\
 & (-9385, \quad 2063285, \quad \cdot \quad -645, \\
 & \quad 8968, \quad -2116448, \quad 645, \pitchfork X, Y, Z, 10) = 0; \\
(1) & (\cdot \quad 3040, \quad -12685, \quad 2242 \pitchfork X, Y, Z, 10) = 0, \\
 & (-3040, \quad \cdot \quad 32065, \quad -8170 \pitchfork X, Y, Z, 10) = 0, \\
 & (12685, \quad -32065, \quad \cdot \quad 10443, \\
 & \quad -2242, \quad 8170, \quad -10443, \pitchfork X, Y, Z, 10) = 0; \\
(3) & (\cdot \quad -484120, \quad 1175, \quad -1064 \pitchfork X, Y, Z, 10) = 0, \\
 & (484120, \quad \cdot \quad 482565, \quad -485184 \pitchfork X, Y, Z, 10) = 0, \\
 & (-1175, \quad -482565, \quad \cdot \quad 117, \\
 & \quad 1064, \quad 485184, \quad -117, \pitchfork X, Y, Z, 10) = 0; \\
(5) & (\cdot \quad 5510, \quad 245, \quad -266 \pitchfork X, Y, Z, 10) = 0, \\
 & (-5510, \quad \cdot \quad 4885, \quad -5776 \pitchfork X, Y, Z, 10) = 0, \\
 & (-245, \quad -4885, \quad \cdot \quad 3, \\
 & \quad 266, \quad 5776, \quad -21, \pitchfork X, Y, Z, 10) = 0; \\
(6) & (\cdot \quad -5320, \quad 375, \quad -456 \pitchfork X, Y, Z, 10) = 0, \\
 & (5320, \quad \cdot \quad 3805, \quad -5776 \pitchfork X, Y, Z, 10) = 0, \\
 & (-375, \quad -3805, \quad \cdot \quad 81, \\
 & \quad 456, \quad 5776, \quad -81, \pitchfork X, Y, Z, 10) = 0.
\end{array}
$$

The several lines intersect as they should do, the coordinates of the points of intersection being as follows

	$1'$	$3'$	4	$5'$	$6'$
1	.	.		$\left(\frac{305}{1064}, -\frac{424171}{8968}\right)$	
			.		$\left(-\frac{4721}{2}, \frac{47}{2}\right)$
2	.	.		$\left(\frac{9693}{1064}, -\frac{2111073}{2484}\right)$	
			.		$\left(\frac{6521}{66}, \frac{24727}{464}\right)$
$2'$	$\frac{9}{1}$		.		
			$\frac{8}{2}$		$\frac{3}{7}$
$4'$		$\frac{47521}{42} + \frac{10801}{283}$	.		
	$\frac{71744}{2128}$		.		
3		.	.	$\left(\frac{24211}{152} + 233, -\frac{1891}{152} + \frac{727}{233}\right)$	
					.

viz. the coordinates of $12'$ (intersection of lines 1 and $2'$) are $\left(\frac{305}{1064}, -\frac{424171}{8968}\right)$, and so in other cases; where there is no divisor the coordinates are integers. I find however, on laying down the figure, that the lines 3 and 4, $3'$ and $4'$ come so close together, that the figure cannot be obtained with any accuracy.

460. NOTE ON MR FROST'S PAPER ON THE DIRECTION OF LINES OF CURVATURE IN THE NEIGHBOURHOOD OF AN UMBILICUS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. x. (1870), pp. 111–113.]

I REMARK as follows:

1. In regard to a quadric surface, it is not, I think, correct to say that the generatrices through an umbilic are curves of curvature; notwithstanding that, as shown p. 80, the normals at every point of such generatrix lie in one plane and consequently intersect. The way in which these generatrices as quasi-curves-of-curvature present themselves is as follows:

The curves of curvature satisfy a certain differential equation, the complete integral of which gives these curves as the intersections of the given quadric surface by the series of confocal surfaces

$$\frac{x^2}{a^2+\theta}+\frac{y^2}{b^2+\theta}+\frac{z^2}{c^2+\theta}=1,$$

θ being the constant of integration of the differential equation. The singular solution of the differential equation, or envelope of the curves of curvature determined as above, gives the umbilicar generatrices.

2. In regard to a surface in general, I think it must be considered, not that there pass through the umbilic three distinct curves, but that the umbilicar curve of curvature is a curve having at the umbilic a triple point, or rather a point at which there are in general three distinct directions of the curve. The umbilicar curve of curvature in fact presents itself as the curve belonging to a certain value of the constant of integration h; in order that the curve of curvature may pass through a given point on the surface, h must satisfy a certain quadratic equation, that is for a given point of the surface there are two values of h, and therefore two curves of curvature: but an umbilic is a point for which (as in effect shown, p. 81, for the particular case of a quadric surface) the two values of h become equal; that is, there is through the umbilic only a singular curve of curvature; but $\frac{dh}{dx}$ is determined by a cubic equation, and the umbilic is thus (as just mentioned) a point at which there are in general three distinct directions of the curve.

3. Some researches on the subject are contained in my paper "On Differential Equations and Umbilici," *Phil. Mag.*, vol. XXVI. (1863), pp. 373–379 and 441–452, [330]. It is noticeable that in the integral equations which I have there obtained for the differential equations $cy(\phi^2-1)+(a-c)x\phi=0$, and the more general form $(bx+cy)(\phi^2-1)+2(fx+gy)\phi=0$, which belong to the neighbourhood of an umbilic, the curve through the umbilic does break up into three distinct curves; and the same is the case with the umbilic on the surface $xyz=1$ presently referred to.

4. In the paper "Mémoire sur les surfaces orthogonales," *Liouv.*, t. XII. (1847), pp. 241–254, M. Serret has given two very remarkable cases of three systems of surfaces intersecting each other at right angles, and consequently in the curves of curvature of the surfaces of each system. It was only on referring to this paper, in connexion with that of Mr Frost, that I perceived an obvious enough simplification of M. Serret's formulae, whereby it appears that the curves of curvature on the surface $xyz=1$ are given as the intersection of this surface with the series of surfaces

$$h=(x^2+\omega y^2)^{\frac{3}{2}}+(x^2+\omega^2 z^2)^{\frac{3}{2}}+(y^2+\omega z^2)^{\frac{3}{2}},$$

where ω is an imaginary cube root of unity; the rationalised equation is of the twelfth order in (x,y,z), and for the particular value $h=0$, reduces itself as is easily seen to

$$0=(y^2-z^2)^2(z^2-x^2)^2(x^2-y^2)^2.$$

The point $x=y=z=1$ is obviously an umbilic on the surface $xyz=1$, and the corresponding value of h being $h=0$, the equation just obtained determines the umbilicar curves of curvature.

Viz. combining therewith the equation $xyz = 1$ of the surface, we have the three hyperbolic curves

$$(x = y,\ xy^3 = 1), \quad (x = \omega y,\ \omega^3 y^4 = 1), \quad (x = \omega^2 y,\ \omega^6 y^4 = 1).$$

461. ON THE GEOMETRICAL INTERPRETATION OF THE COVARIANTS OF A BINARY CUBIC.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. x. (1870), pp. 148–149.]

CONSIDER the binary cubic $U = (a, b, c, d \between x, y)^3$, and its covariants, viz. the discriminant (invariant)

$$\nabla = a^2d^2 - 6abcd + 4ac^3 + 4b^3d - 3b^2c^2,$$

the Hessian

$$H = (ac - b^2)x^2 + (ad - bc)xy + (bd - c^2)y^2,$$

and the cubicovariant

$$\begin{aligned}\Phi = (&\quad a^2d - 3abc + 2b^3)x^3\\ -&(-3abd + 6ac^2 - 3b^2c)x^2y\\ +&(\quad 3acd - 6bd^2 + 3bc^2)xy^2\\ -&(\quad ad^2 - 3bcd + 2c^3)y^3,\end{aligned}$$

connected by the identical equation

$$\Phi^2 - \nabla U^2 = -4H^3.$$

Then if we regard (a, b, c, d) as the coordinates of a point in space, but (x, y) as variable parameters, the equation

$$\nabla = 0$$

represents a quartic torse, having for its cuspidal curve the skew cubic $ac - b^2 = 0$, $ad - bc = 0$, $bd - c^2 = 0$; the equation

$$U = 0$$

is that of the tangent plane to the torse along the line $ax^2+2bxy+cy^2 = 0$, $hax^2+2cxy+dy^2 = 0$: this line meets the cuspidal curve in the point whose coordinates are $a : b : c : d = y^3 : -xy^2 : x^2y : -y^3$. The equation

$$H = 0$$

is that of a quadric cone having the last mentioned point for its vertex, and passing through the cuspidal curve: and the equation

$$\Phi = 0$$

is that of the cubic surface which is the first polar of the same point in regard to the torse.

The equation $\Phi^2 - \nabla U^2 = -4H^3$, writing therein $\nabla = 0$, gives $\Phi^2 = -4H^3$, a result which implies that $U = 0$, $H = 0$ is a certain curve repeated twice, and that $U = 0$, $\Phi = 0$ is the same curve repeated three times. The curve in question is at once seen to be the line of contact $\partial_x U = 0$, $\partial_y U = 0$; it thus appears that the tangent plane $U = 0$ meets the cubic surface $\Phi = 0$ in this line taken three times. This can only be the case if the equation $\Phi = 0$ be expressible in the form $MU + (\partial_x U)^2 = 0$, or, what is the same thing,

$$MU + (\alpha\partial_x U + \beta\partial_y U)^2 = 0,$$

α and β constants, M a quadric function of (a, b, c, d); that is, Φ must be equal to a function of the form

$$MU + (\alpha\partial_x U + \beta\partial_y U)^2.$$

Seeking for this expression of Φ, and writing the symbols out at length, I find that the required identical equation is

$$\begin{aligned}
&(\ \ a^2d-3abc+2b^3)x^3\\
&\qquad\qquad\qquad\qquad -(-3abd+6ac^2-3b^2c)x^2y\\
&\qquad\qquad\qquad\qquad +(\ \ 3acd-6bd^2+3bc^2)xy^2\\
&\qquad\qquad\qquad\qquad -(\ \ ad^2-3bcd+2c^3)y^3\\
&\qquad\qquad\qquad\qquad -(\alpha x-\beta y)^2(a,b,c,d\pitchfork x,y)^2\\
&\qquad\qquad +2\{\alpha(ax^2+2bxy+cy^2)+\beta(bx^2+2cxy+dy^2)\}^2=
\end{aligned}$$

$$(a,b,c,d\pitchfork x,y)^2\cdot\begin{pmatrix} 2a^2 & 6ab & 6b^2 & ad+3bc\\ 6ab & 12ac+6b^2 & 3ad+15bc & 6c^2\\ 6b^2 & 3ad+15bc & 12bd+6c^2 & 6cd\\ ad+3bc & 6c^2 & 6cd & 2d^2\end{pmatrix}\pitchfork(x,y)^2(\alpha,\beta)^2,$$

(where the $\pitchfork$ indicates that the binomial coefficients are not to be inserted, viz. the function on the right hand is $\{2a^2x^2+6abx^2y+6b^2x^2y^2+(-ad+3bc)y^2\}\alpha^2+\&c.$). As a verification remark that for $x=\alpha$, $y=\beta$, the equation becomes simply $2U^2=2U\cdot 2U$.

462. A NINTH MEMOIR ON QUANTICS.

[From the *Philosophical Transactions of the Royal Society of London*, vol. CLXI. (for the year 1871), pp. 17–50. Received April 7,—Read May 19, 1870.]

It was shown not long ago by Professor Gordan that the number of the irreducible covariants of a binary quantic of any order is finite (see his memoir "Beweis dass jede Covariante und Invariante einer binären Form eine ganze Function mit numerischen Coefficienten einer endlichen Anzahl solcher Formen ist," *Crelle*, t. LXIX. (1869), Memoir dated 8 June 1868), and in particular that for a binary quintic the number of irreducible covariants (including the quintic and the invariants) is $= 23$, and that for a binary sextic the number is $= 26$. From the theory given in my "Second Memoir on Quantics," *Phil. Trans.*, 1856, [141], I derived the conclusion, which, as it now appears, was erroneous, that for a binary quintic the number of irreducible covariants was infinite. The theory requires, in fact, a modification, by reason that certain linear relations, which I had assumed to be independent, are really not independent, but, on the contrary, linearly connected together: the interconnexion in question does not occur in regard to the quadric, cubic, or quartic and for these cases respectively the theory is true as it stands; for the quintic the interconnexion first presents itself in regard to the degree 8 in the coefficients and order 14 in the variables, viz. the theory gives correctly the number of covariants of any degree not exceeding 7, and also those of the degree 8 and order less than 14 but for the order 14 the theory as it stands gives a non-existent irreducible covariant $(a, \ldots \between x, y)^{14}$, viz. we have, according to the theory, $5 = (10 - 6) + 1$, that is, of the form in question there are 10 composite covariants connected by 6 syzygies, and therefore equivalent to $10 - 6, = 4$ asyzygetic covariants: but the number of asyzygetic covariants being $= 5$, there is left, according to the theory, 1 irreducible covariant of the form in question. The fact is that the 6 syzygies being interconnected and equivalent to 5 independent syzygies only, the composite covariants are equivalent to $10 - 5, = 5$, the full number of the asyzygetic covariants. And similarly the theory as it stands gives a non-existent irreducible covariant $(a, \ldots \between x, y)^{16}$. The theory being thus in error, by reason that it omits to take account of the interconnexion of the syzygies, there is no difficulty in conceiving that the effect is the introduction of an infinite series of non-existent irreducible covariants, which, when the error is corrected, will disappear, and there will be left only a finite series of irreducible covariants.

Although I am not able to make this correction in a general manner so as to show from the theory that the number of the irreducible covariants is finite, and so to present the theory in a complete form, it nevertheless appears that the theory can be made to accord with the facts; and I reproduce the theory, as well to show that this is so as to exhibit certain new formulae which appear to me to place the theory in its true light. I remark that although I have in my Second Memoir considered the question of finding the number of irreducible covariants of a given degree θ in the coefficients but of any order whatever in the variables, the better course is to separate these according to their order in the variables, and so consider the question of finding the number of the irreducible covariants of a given degree θ in the coefficients, and of a given order n in the variables. (This is, of course, what has to be done for the enumeration of the irreducible covariants of a given quantic and what is done completely for the quadric, the cubic, and the quartic, and for the quintic up to the degree 6 in my Eighth Memoir, *Phil. Trans.* 1867, [405].) The new formulae exhibit this separation thus (Second Memoir, No. 49), writing a instead of x, we have; for the quadric the expression

$$\frac{1}{(1-a)(1-a^2)},$$

showing that we have irreducible covariants of the degrees 1 and 2 respectively, viz. the quadric itself and the discriminant: the new expression is

$$\frac{1}{1-aa^2} \cdot \frac{1}{1-a^2},$$

showing that the covariants in question are of the actual forms $(a, \ldots \pitchfork x, y)^2$ and $(a, \ldots)^2$ respectively. Similarly for the cubic, instead of the expression No. 55,

$$\frac{1}{(1-a)(1-a^2)(1-a^3)(1-a^4)},$$

we have

$$\frac{1-aa^2}{(1-aa^2)(1-ax)(1-ax^{-1})(1-a^4)},$$

exhibiting the irreducible covariants of the forms $(a, \ldots \pitchfork x, y)^3$, $(a, \ldots \pitchfork x, y)^2$, $(a, \ldots)^4(x, y)^3$, and $(a, \ldots)^4$, connected by a syzygy of the form $(a, \ldots)^2(x, y)^2$; and the like for quantics of a higher order.

In the present Ninth Memoir I give the last-mentioned formulae; I carry on the theory of the quintic, extending the Table No. 82 of the Eighth Memoir up to the degree 8, calculating all the syzygies, and thus establishing the interconnexions in virtue of which it appears that there are really no irreducible covariants of the forms $(a, \ldots)^8(x, y)^{14}$ and $(a, \ldots)^8(x, y)^{16}$. I reproduce in part Gordan's theory so far as it applies to the quintic, and I give the expressions of such of the 23 covariants as are not given in my former memoirs; these last were calculated for me by Mr W. Barrett Davis, by the aid of a grant from the Donation Fund at the disposal of the Royal Society. [The expressions referred to are in fact printed, 143.] The paragraphs of the present memoir are numbered consecutively with those of the former memoirs on Quantics.

Article Nos. 328 to 332. Reproduction of my original Theory as to the Number of the Irreducible Covariants.

328. I reproduce to some extent the considerations by which, in my Second Memoir on Quantics, I endeavoured to obtain the number of the irreducible covariants of a given binary quantic $(a, b, \ldots \between x, y)^n$.

Considering in the first instance the covariants as functions of the coefficients $(a, b, c, \ldots)$, without regarding the variables (x, y), and attending only to the following properties—

1°. a covariant is a rational and integral homogeneous function of the coefficients;

2°. if P, Q, R, ... are covariants, any rational and integral function $F(P, Q, R, \ldots)$, homogeneous in regard to the coefficients, is also a covariant, we say—

that the covariants X, Y, ... of the same degree in regard to the coefficients, and not connected by any identical equation $\alpha X + \beta Y \ldots = 0$ (where α, β, ... are quantities independent of the coefficients $(a, b, c, \ldots)$), are asyzygetic covariants, and that a covariant not expressible as a rational and integral function of covariants of lower degrees is an irreducible covariant; and it is assumed that we know the number of the asyzygetic covariants of the degrees 1, 2, 3, ...; say, these are A_1, A_2, A_3, ..., or, what is the same thing, that the number of the asyzygetic covariants of the degree θ, or form $(a, b, \ldots)^\theta$, is equal to the coefficient of a^θ in a given function

$$\Phi(a) = 1 + A_1 a + A_2 a^2 + \ldots + A_\theta a^\theta + \ldots,$$

where I have purposely written a, as a representative of the coefficients $(a, b, c, \ldots)$, in place of the x of my Second Memoir.

329. The theory was, that determining κ_1, κ_2, ... by the conditions

$$\begin{aligned} A_2 &= \kappa_1 \tfrac{1}{2}(\kappa_1 + 1) + \kappa_2, \\ A_3 &= \kappa_1 \tfrac{1}{2}(\kappa_1 + 1)(\kappa_1 + 2) + \kappa_1\kappa_2 + \kappa_3, \end{aligned}$$

that is, throwing

$$1 + A_1 a + A_2 a^2 + A_3 a^3 + \ldots$$

into the form

$$(1-a)^{-\kappa_1}(1-a^2)^{-\kappa_2}(1-a^3)^{-\kappa_3}\ldots,$$

the index κ_r would express the number of irreducible covariants of the degree r less the number of the (irreducible) linear relations, or syzygies, between the composite or non-irreducible covariants of the same degree. Thus $\kappa_2 = \kappa_2$, there would be κ_2 covariants of the degree 2; these give rise to $\frac{1}{2}\kappa_2(\kappa_2 + 1)$ composite covariants of the degree 2; or, assuming that these are connected by k_2 syzygies, the number of asyzygetic composite covariants of the degree 2 would be $\frac{1}{2}\kappa_2(\kappa_2+1) - k_2$; and thence there would be $A_2 - \frac{1}{2}\kappa_2(\kappa_2 + 1) + k_2$, that is, $\kappa_2 + k_2$ irreducible covariants of the same degree; so that (irreducible invariants less syzygies) $(\kappa_2 + k_2) - k_2$ is $= \kappa_2$.

330. The κ_2 syzygies are here irreducible syzygies; for, calling P, Q, R, ... the covariants of the degree 1, there is no identical relations between the terms P^2, Q^2, PQ, ...: imagine for a moment that we could have L such identical relations (viz. this might very well be the case if instead of the $\frac{1}{2}\kappa_1(\kappa_1+1)$ functions P^2, Q^2, PQ, ..., we were dealing with the same number of other quadric functions of these quantities), that is, relations not establishing any relation between P^2, Q^2, PQ, ..., and besides these k_2 non-identical relations as above; then the number of irreducible invariants would be $\kappa_2 + k_2 + L$, and the number of irreducible syzygies being as before k_2, the difference would be not κ_2, but $\kappa_2 + L$. The L identical relations are here relations between composite covariants, and the effect (if any such relation could subsist) would, it appears, be to increase κ_2; between syzygies such identical relations do actually exist, and the effect is contrariwise to diminish the κ; we may, for instance, for the degree s have irreducible covariants less irreducible syzygies $= \kappa_s - L_s$.

331. Assume for a moment that, for a given value of s, κ_s is positive; but for the term L, it would of course follow that there was for the degree in question a certain number of irreducible covariants and it was in this manner that I was led to infer that the number of the covariants of a quintic was infinite—viz. the transformed expression for the number of asyzygetic covariants is

$$= \text{coeff. } a^n \text{ in } (1-a)^{-\kappa_1}(1-a^2)^{-\kappa_2}(1-a^3)^{-\kappa_3}(1-a^4)^{-\kappa_4}\ldots,$$

a product which does not terminate, and as to which it is also assumed that the series of negative indices does not terminate.

332. The principle is the same, but the discussion as to the number of the irreducible covariants becomes more precise, if we attend to the covariants as involving not only the coefficients $(a, b, \ldots)$ but also the variables (x, y); we have then to consider the covariants of the form $(a, b, \ldots)^\theta(x, y)^n$, or, say, of the form $a^\theta x^n$ (degree θ and order n), and the number of the asyzygetic covariants of this form is given as the coefficient of $a^\theta x^n$ in a given function of (a, x), (I write a instead of the z of my Second Memoir in the formulae which contain x and z): by taking account of the composite covariants and syzygies, we successively determine, from the given number of asyzygetic covariants for each value of θ and n, the number of the irreducible covariants for the same values of θ and n. This is, in fact, done for the quintic in my Eighth Memoir up to the covariants and syzygies of the degree 6. But before resuming the discussion for the quintic, I will consider the preceding cases of the quadric, the cubic, and the quartic.

Article Nos. 333 to 336. New formulae for the number of Asyzygetic Covariants.

333. For the quadric $(a, b, c \between x, y)^2$, the number of asyzygetic covariants $a^\theta x^n$ is

$$= \text{coeff. } a^\theta x^{n-2\mu} \text{ in } \frac{1-x}{(1-a)(1-ax)(1-ax^2)},$$

(see Second Memoir, No. 35, observing that q is there $= \theta - \frac{1}{2}\mu$, and that the subtraction of successive coefficients is effected by means of the factor $1-x$ in the numerator. See also Eighth Memoir, No. 251, where a like form is used for the quintic). Writing ax^2 for a, and $\frac{1}{x}$ for x, this is

$$= \text{coeff. } a^\theta x^n \text{ in } \frac{1}{\frac{1}{x}(1-ax^2)(1-a)(1-\frac{1}{x})}.$$

The development is

$$\begin{aligned} &1 \\ &+ ax(x^2+1) \\ &+ a^2(x^4+x^2+1) \\ &+ a^3(x^6+x^4+x^2+1) \\ &\quad \cdots\cdots\cdots\cdots\cdots\cdots \end{aligned}$$

which is

$$= A(x) + A\left(\frac{1}{x}\right),$$

where

$$A(x) = \frac{1}{(1-ax^2)(1-a)(1-\frac{1}{x})},$$

and, since $A(\frac{1}{x})$ contains only negative powers, the required number is

$$= \text{coeff. } a^\theta x^n \text{ in } \frac{1}{(1-ax^2)(1-a)} = \frac{1}{1-aa^2} \cdot \frac{1}{1-a^2},$$

indicating that the covariants are powers and products of (ax^2 and a^2), the quadric itself, and the discriminant. Compare Second Memoir, No. 49, according to which, writing therein a for x, the number of asyzygetic covariants is

$$= \text{coeff. } a^\theta \text{ in } \frac{1}{(1-a)(1-a^2)}.$$

334. For the cubic $(a, b, c, d \between x, y)^3$ the number of asyzygetic covariants $a^\theta x^n$ is

$$= \text{coeff. } a^\theta x^{n-2\mu} \text{ in } \frac{1-x}{(1-a)(1-ax)(1-ax^2)(1-ax^3)},$$

or transforming as before, this is

$$= \text{coeff. } a^\theta x^n \text{ in } \frac{1-x^{-1}}{(1-ax^3)(1-ax^2)(1-ax)(1-ax^{-1})}.$$

the function is here

$$A(x) + A\left(\frac{1}{x}\right),$$

where

$$A(x) = \frac{1 - ax^2}{(1 - ax^3)(1 - ax^2)(1 - ax)(1 - a^4)},$$

(that this is so may be easily verified); and since the second term contains only negative powers, the required number is = coeff. $a^\theta x^n$ in $A(x)$. The formula, in fact, indicates that the covariants are made up of (ax^3, ax^2, ax^4, a^4), the cubic itself, the Hessian, the cubicovariant, and the discriminant, these being connected by a syzygy (a^2x^6) of the degree 6 and order 6. Compare Second Memoir, No. 50, according to which, the number of covariants of degree θ is

$$= \text{coeff. } a^\theta \text{ in } \frac{1 - a^6}{(1 - a)(1 - a^2)(1 - a^3)(1 - a^4)}.$$

335. For the quartic $(a, b, c, d, e \pitchfork x, y)^4$ the number of asyzygetic covariants $a^\theta x^n$ is

$$= \text{coeff. } a^\theta x^{n-2\mu} \text{ in } \frac{1 - x}{(1 - a)(1 - ax)(1 - ax^2)(1 - ax^3)(1 - ax^4)},$$

or transforming as before, this is

$$= \text{coeff. } a^\theta x^n \text{ in } \frac{1 - x^{-1}}{(1 - ax^4)(1 - ax^3)(1 - ax^2)(1 - ax)(1 - ax^{-1})}.$$

the function is here

$$A(x) + A\left(\frac{1}{x}\right),$$

where

$$A(x) = \frac{1 - ax^2}{(1 - ax^4)(1 - ax^3)(1 - ax^2)(1 - ax)(1 - a^5)(1 - a^6)(1 - a^2x^2)},$$

and the second term containing only negative powers, the required number is = coeff. $a^\theta x^n$ in $A(x)$. The formula indicates that the covariants are made up of $(ax^4, ax^2, a^2, a^3, a^2x^4)$, the quartic itself, the Hessian, the quadrinvariant, the cubinvariant, and the cubicovariant, these being connected by a syzygy (a^3x^{12}) of the degree 6 and order 12. Compare Second Memoir, No. 51, according to which the number of covariants of degree θ is

$$= \text{coeff. } a^\theta \text{ in } \frac{1 - a^6}{(1 - a)(1 - a^2)(1 - a^3)(1 - a^4)(1 - a^5)}.$$

336. For the quintic $(a, b, c, d, e, f \pitchfork x, y)^5$ the number of asyzygetic covariants $a^\theta x^n$ is

$$= \text{coeff. } a^\theta x^{n-2\mu} \text{ in } \frac{1 - x}{(1 - a)(1 - ax)(1 - ax^2)(1 - ax^3)(1 - ax^4)(1 - ax^5)},$$

or transforming as before, this is

$$= \text{coeff. } a^\theta x^n \text{ in } \frac{1 - x^{-1}}{(1 - ax^5)(1 - ax^4)(1 - ax^3)(1 - ax^2)(1 - ax)(1 - ax^{-1})}.$$

The developed expression is

$$\begin{aligned}
&\frac{1}{x}\\
&+aa^2\\
&+a^2(x^4+x^2+1)\\
&+a^3(x^{-6}+x^{-4}+x^{-2})\\
&\quad\cdots\cdots\cdots\cdots\cdots\cdots
\end{aligned}$$

but here there is not any finite function $A(x)$ such that this development is

The numerical coefficients are of course the same as those in the development of the untransformed function; viz. they are the numbers given in the third column of Table No. 82 (Eighth Memoir), and also (carried further) in the third column of the following Table, No. 87. And we can, from the discussion of these coefficients, deduce the form of $A(x)$, viz. this is

$$\begin{array}{ccc}
\frac{1}{1-a^2x^{14}} & \frac{1}{1-a^2x^{13}} & \frac{1}{(1-a^2x^{10})^2}\\[6pt]
\frac{1}{1-a^2x^{12}} & \frac{1}{1-a^2x^{11}} & \frac{1}{(1-a^2x^{8})^2}\\[6pt]
\frac{1}{1-a^2x^{10}} & \frac{1}{(1-a^2x^{9})^2} & \frac{}{(1-a^2x^{6})^2}\\[6pt]
\frac{1}{1-a^2x^{8}} & \frac{1}{1-a^2x^{7}} & \\[6pt]
\frac{1}{1-a^2x^{6}} & &
\end{array}$$

$$\begin{array}{cccccc}
\frac{1}{1-ax^2} & \frac{1}{1-ax^5} & \frac{1}{1-a^2x^4} & \frac{1}{1-a^2x^3} & \frac{1}{1-a^2x^2} & \ldots\\[10pt]
2 & 5 & 4 & 3 & 2 & \\
 & & 3 & 1 & & 20\\
 & & & & & 14\\
 & & & & & 8
\end{array}$$

where, for shortness, I have written $1-a^2x^{14}$ to stand for $(1-a^2x^{14})(1-a^2x^{13})$, and so in other cases; moreover in the third column of the numerator the $(9)^2$ shows that the factor is $(1-a^2x^9)^2$, and so in other cases: this will be further explained presently.

Compare herewith the form, Second Memoir, No. 52, viz. the number of asyzygetic covariants of the degree θ is

$$=\text{coeff. } a^\theta \text{ in } (1-a)^{-\kappa_1}(1-a^2)^{-\kappa_2}(1-a^3)^{-\kappa_3}(1-a^4)^{-\kappa_4}(1-a^5)^{-\kappa_5}(1-a^6)^{-\kappa_6}(1-a^7)^{-\kappa_7}(1-a^8)^{-\kappa_8}\ldots,$$

each index being, it will be observed, equal to the number of factors in the numerator, less the number of factors in the denominator, in the corresponding column of the new formula.

Article Nos. 337 to 346. The 23 Fundamental Covariants.

337. Gordan's result is that the number of the irreducible covariants of the binary quintic is $= 23$. I represent these by the letters A, B, C, ..., W, identifying such of them as were given in my former Memoirs on Quantics with the Tables of these Memoirs, and the new ones, O, P, R, S, T, V, with the Tables Nos. 90, 91, 92, 93, 94, 95 of the present Memoir.

Table No. 87. Identification of the 23 irreducible covariants of the binary quintic.

			Table No.
A	$= (a, b, c, d, e, f \pitchfork x, y)^5$	$f = (f)^1$	13
B	$= \frac{1}{12}(A, A)^2$	$(\pitchfork x, y)^2\ \phi = (f, f)^2$	14
C	$= \frac{1}{12}(A, B)^1$	$(\pitchfork x, y)^3\ \psi = (f, \phi)^1$	15
D	$= -\frac{1}{2}(A, B)^2$	$(\pitchfork x, y)^1\ i = (f, \phi)^2$	16
E	$= \frac{1}{4}(A, C)^2$	$(\pitchfork x, y)^2\ (f, \psi)^2$	17
F	$= \frac{1}{2}(A, C)^3$	$(\pitchfork x, y)^1\ (f, \psi)^3$	18
G	$= -\frac{1}{4}(B, B)^2$	$(\pitchfork x, y)^2\ \xi = (\phi, \phi)^2$	19
H	$= -\frac{1}{2}(B, C)^2 + \frac{1}{4}B^2$	$(\pitchfork x, y)^2\ \rho = (\phi, \psi)^2$	20
I	$= -\frac{1}{2}(B, C)^3$	$(\pitchfork x, y)^1\ (\phi, \psi)^3$	21
J	$= -\frac{1}{2}(B, D)^2$	$(\pitchfork x, y)^1\ \sigma = (j)^2$	22
K	$= \ \ (B, D)^3$	$(\pitchfork x, y)^1\ (j)^3$	23
L	$= -\frac{1}{2}(A, H)^3 + \frac{1}{4}BE$	$(\pitchfork x, y)^1\ (f\rho)^3$	24
M	$= -\frac{1}{12}(B, H)^2 - \frac{1}{4}BG$	$(\pitchfork x, y)^2\ \tau = (\rho, \psi)^2$	83
N	$= \ \ (B, H)^3$	$(\pitchfork x, y)^1\ (\rho)^3$	84
O	$= - \ \ (B, J)^2$	$(\pitchfork x, y)^1\ (\sigma)^2$	*90
P	$= -\frac{1}{2}(A, M)^3 - BK$	$(\pitchfork x, y)^1\ (f\tau)^3$	*91
Q	$= \frac{1}{12}(B, M)^2$	$(\pitchfork x, y)^2\ (\tau)^2$	25
R	$= -\frac{1}{2}(B, M)^3$	$(\pitchfork x, y)^1\ (\tau)^3$	*92
S	$= -96(D, M)^2 + 160J - 7GK$	$(\pitchfork x, y)^1\ (\tau)^1$	*93
T	$= - \ \ (J, M)^2$	$(\pitchfork x, y)^1\ \gamma = (j\tau)^2$	*94
U	$= \ \ (J, O)^2 + \frac{1}{4}OQ$	$(\pitchfork x, y)^1\ ((\sigma)^2, \psi)^2$	29
V	$= - \ \ (B, T)^2$	$(\pitchfork x, y)^1\ (\xi\gamma)^2$	*95
W	$= -\frac{1}{2}(O, T)^2$	$(\pitchfork x, y)^1\ ((\sigma)^2, \gamma)^2$	29α

338. The Table exhibits the generation of the several covariants; viz. (A, B) denotes $\partial_x A \cdot \partial_y B - \partial_y A \cdot \partial_x B$, $(A, B)^2$ denotes $\partial_x^2 A \cdot \partial_y^2 B - 2\partial_x A \partial_y A \cdot \partial_x B \partial_y B + \partial_y^2 A \cdot \partial_x^2 B$, &c. (see post. No. 348). The column f, $i = (ff)^2$, &c. shows Gordan's notation, and the generation of his 23 forms ($(ff)^2$ written as with him for $(f, f)^2$, &c.): it will be observed that the forms are not identical; if the calculations had been made de novo, I should have adopted his values, simply omitting numerical factors of the several forms (thus every term of t, $= (ff)^4$ contains the factor $2 \cdot (120)^2 = 28800$): of course the presence of these numerical factors renders the f, t, ψ, &c. as they stand inconvenient for the expression of results; and the numerical fixation of the values was no part of Gordan's object. But by reason of the existing Tables the change of notation is in fact more than this; thus H instead of being a submultiple of $(B, C)^2$, that is, of $\frac{1}{2}\psi$, is in fact $= -\frac{1}{2}(B, C)^2 + \frac{1}{4}B^2$; and so in other cases. If the occasion for it arises, there is no difficulty in expressing any one of the forms f, t, ψ, &c. in terms of the $(A, B, C \ldots V, W)$; thus in the instance just referred to, $\rho = (\phi C)^2$, we have

$$\phi = (\phi)^2 = (\psi, A)^2 = 800C,$$

and

$$t = \tfrac{1}{2}(ff)^4 = \tfrac{1}{2}(\psi, A)^4 = 2 \cdot 120 \cdot 8B,$$

whence $\rho = 2304000\,(B, C)^2$; also $(B, C)^2 = -5H + 2R$; and therefore, finally,

$$2J = -11520000\,H + 4608000\,B^2.$$

339. I remark upon the value $S = -96(D, M)^2 + 160J - 7GK$, that S is the complete value of a covariant $(\pitchfork\, x, y)^1(\pitchfork\, x, y)^1$, the leading coefficient of which is given in Table No. 86 of my Eighth Memoir; the form (D, M), omitting a numerical factor (if any), would have had smaller numerical coefficients, but there is in the form actually adopted the advantage that it vanishes for $a = 0$, $b = 0$, that is, when the quintic has two equal roots, [see post. No. 346].

340. I now form the following Table No. 88, viz. this is the Table No. 82 of my Eighth Memoir, carried as far as a^8, but with the composite covariants expressed by means of the foregoing letters A, B, C, $\ldots$, W; instead of giving the syzygies as in Table No. 82, I transfer them to a separate Table, No. 89. In all other respects the arrangement is as explained, Eighth Memoir, No. 253; but in place of N, S, S' I have written $\aleph$, ζ, ζ' to denote new covariant, new syzygy, derived syzygy, respectively; and I have, as to the terms a^7x^5, a^8x^2 respectively, introduced the new symbol σ to denote an interconnexion of syzygies, as appearing by the Table No. 89, and as will be further explained.

Table No. 88.

[In this Table and the subsequent Table 89, I have for convenience used, instead of capitals, the small italic letters a, b, c, $\dots w$ to denote the 23 irreducible covariants of the quintic.]

Ind. a.	Ind. x.	Coeff.		
1	5	1	a	$*$
	3			
	1			
2	10	1	a^2	
	8			
	6	1	c	#
	4			
	2	1	b	$*$
3	15	1	a^3	
	13			
	11	1		
	9	1	f	$*$
	7	1	ah	
	5	1	e	$*$
	3	1	d	$*$
	1			

Table No. 88 (continued).

Ind. a.	Ind. x.	Coeff.		
4	20	1	a^4	
	18			
	16	1	a^2c	
	14	1	af	
	12	2	a^2b, c^2	
	10	1	ae	
	8	2	ad, bc	
	6	1	i	*
	4	2	b^2, h	*
	2			
		1	g	*
5	25	1	a^5	
	23			
	21	1	a^3c	
	19	1	a^2f	
	17	2	a^2b, ae^2	
	15	2	a^2e, cf	
	13	2	a^2d, abc	
	11	2	ai, bf, ce	ζ
	9	3	ah^2, ah, cd	
	7	2	bc, l	#
	5	2	ag, bd	
	3	1	k	
	1	1	j	$\aleph$

Table No. 88 (concluded).

Ind. a.	Ind. x.	Coeff.		
8	40	1	a^8	
	38			
	36	1	a^6c	
	34	1	a^5f	
	32	2	a^4b, a^2c^2	
	30	2	a^4e, a^3cf	
	28	3	a^4d, a^3bc, ac^2, a^2f^2	ζ'
	26	3	a^3i, a^2bf, a^2ce, ac^2f	ζ'
	24	5	a^3b^2, a^3h, a^2cd, a^2bc^2, a^2ef, cf^2, c^3	ζ'
	22	4	a^3be, a^2l, a^2ci, a^2df, $abcf$, ahe, c^2e	$2\zeta'$
	20	6	a^3g, a^2bd, a^2b^2, a^2ch, a^2e^2, ac^2d, afi, be^2, bf^2, cef	σ
	18	5	a^2j, a^2bi, a^2de, ab^2f, $abce$, adi, afh, c^3, cdf	$4\zeta'$
	16	7	a^2k, a^2b^2, a^2bh, a^2cg, a^2d^2, $abcd$, aei, ab^2c, bef, c^2i, ce^2, fl	$5\zeta'$
	14	5	a^2n, ab^3, abl, ack, adi, aeh, afg, bci, bdf, cde	σ
	12	7	a^2bg, abn, ab^2l, acj, adh, b^4, b^2ch, bc^2, c^2k, cd^2, el, fk, i^2	3ζ
	10	5	abk, aeg, ap, b^2i, bde, cn, dl, fj, hi	3ζ
	8	6	abj, adg, b^3, b^2h, beg, bd^2, cm, ek, h^2	2ζ
	6	3	ao, bn, dk, ej, gi	2ζ
	4	4	ifg, bm, dj, gh	#
	2	1	r	*
	0	2	f^2, q	*

Table No. 89.

[See ante Table No. 88.]

(5, 11)	$ai + bf - ce = 0$
(6, 18)	$c^2d - a^2bc + ie^2 + l^2 = 0$
(6, 14)	$a^2h - bacd - ibc^2 - ef = 0$
(6, 12)	$al - 2ci + 3df = 0$
(6, 10)	$a^2g - 4abd - 4b^3 - e^2 = 0$
(6, 8)	$ok + 2bi - 3de = 0$
(6, 6)	$aj - b^2 + 2bh - cd - g^2 = 0$
(7, 15)	$a^2bd - a^2bc + ach - b^2d - fi = 0$
(7, 13)	$a^2k - abi - 3bg + bci + 3fh = 0$
(7, 11)	$a^2j - ab^2 + abh - 9ad^2 - 6bcd - ei = 0$
(7, 9)	$an - b^2e - bdi + 2eh - fg = 0$
	$m + bdi - eh + fg = 0$
	$2ck - bdi + eh - fg = 0$
(7, 7)	$am + 2b^2l + cj - Mh = 0$

Table No. 89 (continued).

$$
\begin{array}{rl}
\sigma,\ (8,\ 20) & a^2(a^2g - 12abd - 4b^3 - e^2) \quad \text{repr. } (6,\ 10) \\
& -\, a(a^2bd - abe^2 + cch - 6c^2d - fi) \quad \text{repr. } (7,\ 15) \\
& +\, b(a^2d - a^2bc + ic^2 + l^2) \quad \text{repr. } (6,\ 18) \\
& +\, c(a^2h - bacd - ibc^2 - ef) \quad \text{repr. } (6,\ 14) \\
& -\, f(ai + bf - ce) = 0 \quad \text{repr. } (5,\ 11)
\end{array}
$$

$$
\begin{array}{rl}
\sigma,\ (8,\ 14) & a(an - b^2e - bdi + 2eh - fg) \quad \text{repr. } (7,\ 9) \\
& +\, a(2bl + bdi - eh + fg) \quad \text{repr. } (\quad,\ ) \\
& +\, a(2ck - 2di + eh - fg) \quad \text{repr. } (\quad,\ ) \\
& -\, 2b(al - 2ci + 3df) \quad \text{repr. } (6,\ 12) \\
& -\, 2c(ok + 2bi - 3de) \quad \text{repr. } (6,\ 8) \\
& +\, 6d(ai + bf - ce) = 0 \quad \text{repr. } (5,\ 11)
\end{array}
$$

$$
\begin{array}{rl}
(8,\ 12) & ab^2d - b^2c + 2bch - c^2g + i^2 = 0 \\
& \quad -\, 3adi - 2bch + 2c^2g + 18cd^2 + fk - 2i^2 = 0 \\
& \quad el + fk - 2i^2 = 0 \\
(8,\ 10) & abk - cn - bl - 2fj + hi = 0 \\
& \quad ap + 2cm + lj = 0 \\
& \quad b^2i + cn + 3dl + fj - 2hi = 0 \\
(8,\ 8) & abj - b^3 + 4b^2h - 9bd^2 + 12cm - ek - 3h^2 = 0 \\
& \quad adg + 2b^2h - 12bd^2 + 3cm - ek - 2h^2 = 0 \\
(8,\ 6) & ao + bk - 3ej + 2gi = 0 \\
& \quad bn + 3dk - ej + gi = 0
\end{array}
$$

342. In illustration take any one of the lines of Table No. 88, for instance [resuming the notation by capital letters] the line

$$(7,\ 17)\quad 4\quad A^2BE,\ A^2L,\ ACI,\ ADF,\ BCF,\ OE\quad |\quad 2\zeta'\quad |$$

there are here 6 composite covariants, but the number of asyzygetic covariants is $= 4$: there must therefore be $6-4, = 2$ syzygies; we have however (see Table No. 89) two derived syzygies of the right form, viz. these are

$$A(AL-2CI+3DF)=0, \qquad C(AI+BF-CE)=0,$$

which are designated as $2\zeta'$, and there is consequently no new syzygy ζ.

But in the line

$$(7,\ 15)\quad 5\quad A^3G,\ A^2BD,\ AR,\ ACH,\ AE^2,\ CD,\ FI\quad |\quad \zeta',\ \zeta\quad |$$

there are 7 composite covariants, but the number of asyzygetic covariants is $= 5$; there must therefore be $7-5, = 2$ syzygies. One of these is the derived syzygy

$$A(A^2G-E^2-12ABD-4R-G)=0,$$

which is designated by ζ'; the other is a new syzygy (see Table No. 89),

$$A^2BD-ABC^2+ACH-GCD-FI=0,$$

designated by ζ.

343. Take now the line

$$(8,\ 20)\quad 6\quad A^4G,\ A^3BD,\ A^2B^2C,\ A^2CH,\ A^2E^2,\ ACD,\ AFI,\ BG^2,\ BF^2,\ GEF\quad |\quad 5\zeta',\ \sigma\quad |$$

there are here 10 composite covariants, but the number of irreducible covariants is $= 6$; there should therefore be $10-6, = 4$ syzygies. There are, however, the 5 derived syzygies

$$A^2(A^2G-12ABD-4R-G-E^2)=0,\ \text{\&c. (see Table No. 89)}$$

designated by $5\zeta'$; since these are equivalent to 4 syzygies only there must be 1 identical relation between them (designated by σ), viz. this is the equation $= 0$ obtained by adding the several syzygies, multiplied each by the proper numerical factor as shown Table No. 89.

344. Again, for the line

$$(8,\ 14)\quad 5\quad A^3BH,\ AB^2E,\ ABL,\ AGK,\ ADI,\ AEH,\ AFG,\ BGI,\ BDF,\ GDE\quad |\quad 6\zeta',\ \sigma\quad |$$

there are here 10 composite covariants, but only 5 irreducible covariants; there should therefore be $10-5, = 5$ syzygies; we have in fact the 6 derived syzygies

$$A(AN-B^2E-6DI+2Eh-FG)=0,\ \text{\&c. (see Table No. 89)}$$

designated by $6\zeta'$; these must therefore be connected by 1 identical relation (designated by σ), viz. this is the equation $= 0$ obtained by adding the several syzygies, each multiplied by the proper numerical factor as shown Table No. 89.

345. These two cases (σ) are in fact the instances which present themselves where a correction is required to my original theory. The two identical relations in question were disregarded in my original theory, and this accordingly gave the two non-existent irreducible covariants $(a,\ldots \between x,y)^{14}$ and $(a,\ldots \between x,y)^{16}$. And reverting to No. 336, these give in the denominator of $A(x)$ the factors $(1-a^2x^{14})(1-a^2x^{16})$. In virtue hereof, writing $x=1$, we have in $A(x)$ the factor $(1-a)^{-\kappa_1}(1-a^2)^{-\kappa_2}\ldots$ agreeing with the function $(1-a)^{-\kappa_1}(1-a^2)^{-\kappa_2}\ldots(1-a^8)^{-\kappa_8}\ldots$.

And we thus see that the denominator factors of $A(x)$ do not all of them refer to irreducible covariants; viz. we have

aa^2, a^2x^4, a^2x^3, a^2x^2, a^3, a^3x^4, a^3x^2, a^4x^2, a^4, a^4x^6, a^5x^2, a^5x^4, a^6x^2, a^6x^4, a^7x^2, a^8x^2, a^8,

each referring to an irreducible covariant, but a^8x^{14} and a^8x^{16} each referring to an identical relation (σ) or interconnexion of syzygies. And we thus understand how, consistently with the number of the irreducible covariants being finite, the expression for $A(x)$ may be as above the quotient of two infinite products; there will be in the denominator a finite number of factors each referring to an irreducible covariant, but the remaining infinite series of denominator factors will refer each factor to an identical relation or interconnexion of syzygies. But I do not see how we can by the theory distinguish between the two classes of factors, so as to determine the number of the irreducible covariants, or even to make out affirmatively that the number of them is finite.

346. The new covariants O, P, R, S, T, V are as follows:

[Table No. 90 (Covariant O),

Table No. 91 (Covariant P), Table No. 92 (Covariant R), Table No. 93 (Covariant S), Table No. 94 (Covariant T), Table No. 95 (Covariant V),

printed in the paper 143, "Tables of the Covariants M to W of the Binary Quintic: from the Second, Third, Fifth, Eighth, Ninth and Tenth Memoirs on Quantics" with the insertion as therein mentioned of the terms with zero coefficients. The covariant $S, = -96(D,M)^2 + 160J - 7GK$, of the present Memoir is there called S', and there is given the more simple form $S = (D,M)$, of this covariant.]

Article Nos. 347 to 365. Sketch of Professor Gordan's proof for the finite Number, = 23, of the Covariants of a Binary Quintic.

347. I propose to reproduce the leading points of Professor Gordan's proof that the binary quintic $(a, b, c, d, e, f \pitchfork x, y)^5$ has a finite system of 23 covariants, viz. a system such that every other covariant whatever is a rational and integral function of these 23 covariants.

348. **Derivation.** Consider for a moment any two binary quantics ϕ, ψ of the same or different orders, and which may be either independent quantics, or they may be both or one of them covariants, or a covariant, of a binary quantic f. We may form the series of derivatives

$$\begin{aligned}
(\phi, \psi)^0 &= \phi\psi, \\
(\phi, \psi)^1 &= 12\,\phi_x\psi_y - \phi_y\psi_x, \\
(\phi, \psi)^2 &= 12^2\,\phi_{xx}\psi_{yy} - 2\phi_x\phi_y \cdot \psi_x\psi_y + \phi_{yy}\psi_{xx},
\end{aligned}$$

where, however, there is no occasion to use the notation $(\phi, \psi)^0$ (as this is simply the product $\phi\psi$), and the succeeding derivatives may (when there is no risk of ambiguity) be written more shortly $(\phi\psi)$, $(\phi\psi)_2$, $(\phi\psi)_3$, &c.; in all that follows the word "derivative" (Gordan's Uebereinanderschiebung) is to be understood in this special sense.

349. The degree of the derivative $(\phi, \psi)^k$ is the sum of the degrees of the constituents ϕ, ψ; the order of the derivative is the sum of the orders less $2k$; it being understood throughout that the word degree refers to the coefficients, and the word order to the variables. In speaking generally of the covariants or of all the covariants of a quantic f, or of the covariants or all the covariants of a given degree or order, we of course exclude from consideration covariants linearly connected with other covariants (for otherwise the number of terms would be infinite); but unless it is expressly so stated, we do not carry this out rigorously so as to make the system to consist of asyzygetic covariants; viz. it is assumed that the system is complete, but not that it is divested of superfluous terms.

350. **Theorem A.** The covariants of a quantic f of a given degree m can be all of them obtained by derivation from f and the covariants of the next inferior degree $(m-1)$.

In particular for the degree 1 the only covariant is the quantic f itself; for the degree 2 the covariants are $(ff)^2$, $(ff)^4$, $(ff)^6$, …: using for a moment β to denote each of these in succession, the covariants of the third degree are $(\beta f)^0$, $(\beta f)^2$, $(\beta f)^4$, … and so on.

351. Suppose that the covariants of the second degree $(ff)^2$, $(ff)^4$, $(ff)^6$, … are in this order represented by β_1, β_2, β_3…, then the covariants of the third degree written in the order

$$(\beta_1 f)^0,\ (\beta_1 f)^2,\ (\beta_1 f)^4,\ \ldots\ (\beta_2 f)^0,\ (\beta_2 f)^2,\ (\beta_2 f)^4,\ \ldots\ (\beta_3 f)^0,\ (\beta_3 f)^2,\ (\beta_3 f)^4,\ \ldots$$

may be represented by γ_1, γ_2, γ_3, …, the covariants of the fourth degree written in the order

$$(\gamma_1 f)^0,\ (\gamma_1 f)^2,\ (\gamma_1 f)^4,\ \ldots\ (\gamma_2 f)^0,\ (\gamma_2 f)^2,\ (\gamma_2 f)^4,\ \ldots\ (\gamma_3 f)^0,\ (\gamma_3 f)^2,\ (\gamma_3 f)^4,\ \ldots$$

may be represented by δ_1, δ_2, δ_3…, and so on: we thus obtain in a definite order the covariants of a given degree m; say, these are μ_1, μ_2, μ_3, μ_4, …; any term μ_t is said to be a later term than the preceding terms μ_1, μ_2, … and an earlier term than the following ones, μ_t, μ_{t+1}, &c.

Observe that each term μ_t is a derivative $(\lambda, f)^k$, the derivatives of an earlier λ are earlier than those of a later λ; and as regards the derivatives of the same λ, the derivative with a less index of derivation is earlier than that with a greater index of derivation, or, what is the same thing, those are earlier which are of the higher order.

352. The series μ_1, μ_2, μ_3, μ_4, … is not asyzygetic; we make it so, by considering in succession whether the several terms μ_1, μ_2, … respectively are expressible as linear functions of the earlier terms, and by omitting every term which is so expressible.

The reduced series thus obtained is called T_1, T_2, T_3, …. Observe that not every μ is a T, but that every T is a μ; every T therefore arises from a derivation upon f and a certain term λ; which term λ (supposing the λ series reduced in like manner to S_1, S_2, S_3, …) is a linear function of certain of the S's. Each later T is derived from later S's, or it may be from the same S's as an earlier T; viz. if the later T is derived from $(S_1, S_2, \ldots S_a)$, then the earlier T is derived, it may be, from $(S_1, S_2, \ldots S_a)$, or from $(S_1, S_2, \ldots S_{a-k})$, but so that there is not in the series any term later than S_a.

And if, considering any T as thus derived from certain of the S's, and in like manner each of these S's as derived from certain of the R's, and so on, we descend to any preceding series,

$$\mathrm{M}_1,\ \mathrm{M}_2,\ \mathrm{M}_3,\ \ldots$$

it will appear that the T is derived from a certain number $(\mathrm{M}_1, \mathrm{M}_2, \ldots \mathrm{M}_r)$ of the terms of this series.

353. The quadricovariants $(ff)^2$, $(ff)^4$, $(ff)^6$, ... are of different orders, and consequently asyzygetic. They form therefore a series such as the T-series, and they may be represented by

$$\mathrm{B}_1,\ \mathrm{B}_2,\ \mathrm{B}_3,\ \ldots$$

Supposing f to be of the order n, B_1 is of the order $2n$, B_2 of the order $2n-4$, B_3 of the order $2n-8$, and so on. Those terms which are of an order greater than n, are said to be of the form W (agreeing with a subsequent more general definition of W); those which are of an order equal to or less than n, are said to be of the form X: that is, the earlier terms of the B series are W, and the later terms are X; viz. the X terms taken in order, beginning with the earliest, are X_1, X_2, X_3, ...

354. By what precedes any particular T is derived from certain terms B_1, B_2, ... B_l of the B series. This series, B_1, B_2, ... B_l, may stop short of the terms X; it may include a certain number of them, say X_1, X_2, ... X_k. The terms derived from the X's are in the sequel denoted by P.

355. Every covariant whatever is a form or sum of forms such as

$$\mathrm{T}_1^{\alpha}\,\mathrm{B}_2^{\beta}\,\mathrm{B}_3^{\gamma}\ \ldots\ \mathrm{B}_l^{\lambda}\,\mathrm{P}_1^{\mu}\ \ldots\ \mathrm{P}_t^{\nu};$$

writing in regard to any such expression

$$\Sigma\ \text{ind. } 1 = i,\ \Sigma\ \text{ind. } 2 = j,\ \ldots$$

(viz. i is the sum of all those indices α, β, &c. which belong to a term containing the symbolic number 1, j the sum of all the indices α, γ, &c. which belong to a term containing the symbolic number 2, and so on) then each of the numbers i, j, ... is at most $= n$, that is $n-i$, $n-j$, ... may be any of them $= 0$, but they cannot any of them be negative; the degree of the function is $= m$, and its order is $= mn - i - j \ldots$. It is to be further observed that the form is a function of the differential coefficients of f of the orders $n-i$, $n-j$, &c. respectively. It follows that if $n-i$, $n-j$, ... are none of them $= 0$, the form in question may be obtained from a like form belonging to a quantic f' of the next inferior order $n-1$ by replacing therein the coefficients a', b', ... by $ax+by$, $bx+cy$, &c. respectively: for example, if f denote the cubic function $(a, b, c, d \between x, y)^3$, then the Hessian hereof is $12\, f_x f_y$; the like form in regard to the quadric $f' = (a', b', c' \between x, y)^2$ is $12\, f'_x f'_y$, which is $= a'c' - b'^2$; and substituting herein $ax+by$, $bx+cy$, $cx+dy$ for a', b', c' respectively, we have the Hessian $12\, f_x f_y$ of the cubic. A covariant of f derivable in this manner from a covariant of the next inferior quantic f' is said to be a special covariant.